Toward Physics-Faithful Video Generation:

A Depth-Guided Generator, a Human-Annotated Flaw Benchmark, and an Auditable Multi-Tool Critic

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Text-to-video generators produce clips of high visual quality that nonetheless violate physical common sense—objects interpenetrate, motion ignores gravity, quantities fail to conserve—and often simply fail to depict what the prompt requested. Making generated video *physically faithful* requires three things at once: a generator that can be steered by physical structure rather than appearance alone; a way to measure, reliably and auditably, when a clip is wrong; and ground-truth data about the flaws real generators make. This report presents one component for each.

First, a **physics-guided generator**: a rigid-body simulation is rendered into a per-frame depth (or silhouette) control video and passed to the native control branch of a Wan 2.2-VACE video-diffusion backbone, so that scene geometry and motion are driven by the simulation while the text prompt supplies appearance. Second, a **human-annotated flaw benchmark**: 1,500 AI-generated clips carry human misalignment annotations—each with a severity, the violated prompt fragment and a free-text rationale, and, where present, a time span and a box track over the flagged interval—from which we derive a frozen, physics-focused evaluation set. Third, an **auditable multi-tool critic**: a served vision-language reasoner decomposes a prompt into checkable claims, routes those naming a measurable quantity to a frozen open perception specialist that returns localized evidence only, and combines the outcomes with deterministic rules into a three-valued verdict (*plausible*, *implausible*, *abstain*). The evaluation target is human labels, and no closed or proprietary model produces the critic's verdict.

On the evaluation core, compiling a prompt into a plan and executing it recovers 68.9% of localized flaws against 62.6% for verifying each claim independently, at a slightly higher call count. Scored against human 1–5 ratings on VideoPhy-2 under a purpose-trained judge's own protocol, the critic ranks clips for prompt match better than that judge on the benchmark's hard slice, while that judge is better calibrated to the exact rating a human gave. The intended end state is a loop in which the critic supplies the reward that refines the generator; that loop is future work.

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REPORT STATUS AND SCOPE

Scope. This report describes three components built toward physics-faithful video generation: (1) a generator that renders a physics simulation into photorealistic video through a depth/structure control channel; (2) a human-annotated benchmark of prompt-versus-video flaws; and (3) an auditable multi-tool critic that detects such flaws. The three are designed to compose into a loop in which the critic's judgment refines the generator (section 6).

What is reported. The design of each component; the benchmark's contents and statistics; and a paired comparison of the critic against per-claim verification on the evaluation core (section 7). Generation is assessed qualitatively—by inspection and by adherence to the control signal—with one measured study of when the control may be released. External evaluators, cost-matched comparisons and any result on a held-out set are not yet run.

1 Introduction

1.1 Physical faithfulness as the goal

State-of-the-art text-to-video models render strikingly realistic clips that still break physical common sense and often omit or distort what the prompt asked for. Physics-IQ quantifies the first problem directly: strong visual realism coexists with a large gap in physical understanding on its evaluated models and tasks [38]. Closing that gap is our long-term objective, and it decomposes into three coupled needs.

- **A steerable generator.** A model that follows *physical structure*—the geometry and motion of a scene—rather than reproducing the surface statistics of its training data.
- **A reliable, auditable detector of flaws.** A measurement instrument that says when a clip is wrong *and* shows why, at the level of individual evidence—because that judgment is the reward signal any improvement loop consumes.
- **Ground truth about real flaws.** A benchmark of the mistakes real generators make, annotated by people, against which any detector is measured.

1.2 Three contributions and the loop they form

This report contributes one component for each need and states how they are intended to compose.

1. **A physics-guided video generator** (section 2) that converts a rigid-body simulation into a depth or silhouette control video and drives a Wan 2.2-VACE diffusion backbone's native control branch, separating physical structure (from the simulation) from appearance (from the prompt).
2. **A human-annotated flaw benchmark** (section 3): a 1,500-clip corpus of AI-generated video with localized human misalignment labels, and a frozen physics-focused subset used as the evaluation target.
3. **An auditable multi-tool critic** (section 4) that decomposes a prompt into claims, types each one, routes those asking for a quantity, an interval, an outline or a transcription to an open perception specialist returning evidence only, and combines the evidence deterministically into a three-valued verdict.
   Its distinguishing mechanism is that **the video's own timeline is the wiring** (section 5.5). Measurements are not a flat list dispatched in parallel: one measurement's answer decides whether a second is run, where in the clip it is run, and what its result is allowed to mean. An event's occurrence is settled first and gates whether its timing may speak at all; a located interval becomes the explicit list of frames a later, costlier measurement reads; and the composition rule that turns two measured intervals into an order is arithmetic, with a tolerance, that declines rather than breaks a tie.

The end state (section 6) is a closed loop: the benchmark trains and evaluates the critic, and the critic's verdict becomes the reward that refines the generator (section 6).

2 Physics-Guided Video Generation

2.1 Backbone and control mechanism

The backbone is Wan 2.2-VACE-Fun-14B, a two-expert mixture-of-experts video-diffusion model with a native control branch, run through the `WanVACEPipeline` interface; because the two expert stacks exceed the memory of a single 80 GB accelerator, the experts are CPU-offloaded during generation. Generation takes four inputs: the pre-computed control video, a mask, the text prompt, and one scalar conditioning strength. Nothing about appearance is taken from the simulation: the control video decides where surfaces are and how they move, while the prompt alone decides material, colour, lighting and setting (figure 1).

The mask decides which parts of the frame are held fixed and which are synthesised. Dark mask regions mark content to be preserved as given; bright regions mark content to be generated under the guidance of the control. Inside the backbone, the mask splits the control video into a preserved part and a generated part, which are encoded separately and concatenated into the control latent stream. We pass an *all-white* mask, so the preserved part is empty and every pixel of every frame is synthesised: the depth field is used purely as

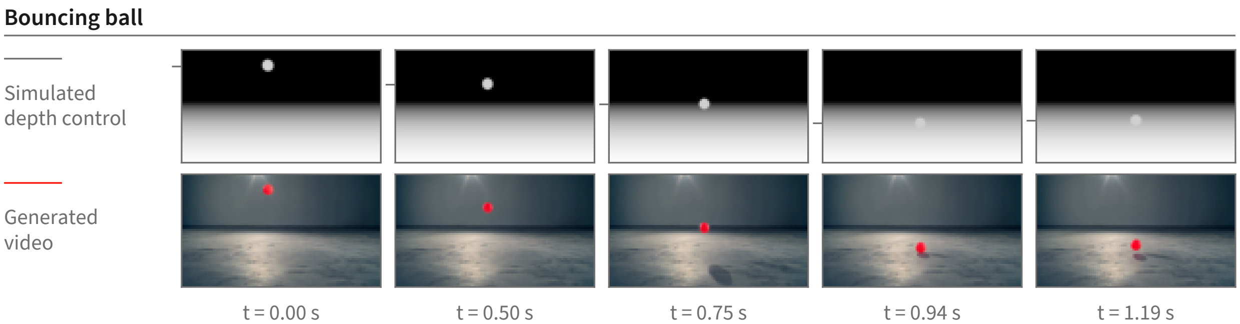


Figure 1 Physics-guided generation, shown on real frames of one bouncing-ball clip. **Top:** the depth control video rendered from the rigid-body simulation, the only geometric input the backbone receives. **Bottom:** the clip generated from it, where the text prompt supplies appearance alone. Both rows are sampled at the same five timestamps of the same 5.06 s, 16 fps, 848×480 clip; the timestamps are chosen by tracking the ball and taking the turning points of its vertical path, so the columns show the drop, the floor contact and the rebound rather than five evenly spaced instants. Tracking the ball independently in each row, the generated path follows the control's with a correlation of 1.0000 horizontally and 0.9997 vertically over all 81 frames, and a median positional disagreement of 0.7 pixels across the frame and 4.7 pixels down. The tick beside each control frame marks the ball's measured centre height; the frames themselves are unaltered.

guidance and no region of it is copied through into the result. This is the mechanism that lets a grey depth render drive a photorealistic clip without tinting it grey.

The conditioning strength is the scalar by which the control latent stream is added into the denoising stream at each of the backbone's control layers. One value applies uniformly to every control layer; a per-layer schedule is also accepted by the interface but is not used here. Turning it down weakens how tightly the generated motion is bound to the simulated motion without changing anything else about the request.

Appearance comes only from text. The prompt names the target scene in the ordinary way ("a bright red rubber ball bouncing on a polished concrete studio floor"), and a single fixed negative prompt—shared by every clip—pushes away from the rendered look the control video literally has, listing cartoon, CGI, render, grayscale and flat shading among the things to avoid. No appearance reference image is supplied. Sampling is deterministic given the seed, and the seed is recorded per clip.

Two properties of this arrangement are worth stating plainly. First, the backbone consumes a control video that was *computed*, not estimated: no monocular depth predictor sits in the loop, so the geometry the generator follows is exactly the simulator's geometry, with no estimation error of its own. Second, this is a control-branch mechanism rather than image-to-image editing: it is neither an SDEdit-style noised re-synthesis [35] of a rendered source nor a copy of source pixels, which is what separates it from conditioning schemes that carry a synthetic source's look into the output.

2.2 Rendering the control signal

One simulation, several registered views. A scene is described as data—a list of moving bodies with initial positions and velocities, a list of static surfaces, and a camera placed by a look-at rule—and the same engine turns any such description into a simulation. Dynamics are integrated in MuJoCo [47] under gravity at 9.81 m/s^2 with a one-millisecond solver step, a fourth-order integrator, and an elliptic friction cone. Output frames are produced on a coarser clock: the requested scene duration is divided into the frame budget, and the solver is advanced by that many sub-steps between frames. At every output frame, and from the *same* solver state, three renderers are queried in turn—a shaded colour view, the depth buffer, and the object-segmentation buffer. Because they read one state, the resulting videos are registered pixel for pixel, so a depth control and a silhouette control for the same scene describe the same geometry and can be substituted for one another without re-simulating.

From depth buffer to control video. The depth buffer holds distance from the camera. It is turned into a control video by an inverted, clipped affine map: distances are mapped so that near surfaces are bright and far surfaces dark, which is the polarity monocular depth estimators produce and therefore the polarity the control branch was trained to read. The mapping's endpoints are robust percentiles—the first and the ninety-seventh—of the depth values in the scene, computed after discarding the far-clip constant that the background sky writes into the buffer, so that a single distant constant cannot compress the whole range into a few grey levels.

Why the mapping is fixed for the whole clip. The endpoints are computed once over all frames of the clip, not per frame. This is a substantive choice, not a detail: it makes the assigned grey level a function of distance alone, so the same distance reads as the same grey in every frame, and a change in brightness therefore means a change in distance. If the endpoints were recomputed per frame—which is the convention monocular depth estimators follow, since they emit one relative map per image—the grey assigned to a distance would depend on the depth distribution of that particular frame. A body holding its distance while other content enters or leaves the view would then change brightness, and the control branch has no way to distinguish that from the body actually moving toward or away from the camera. Fixing the mapping across the clip removes that ambiguity by construction. A per-frame variant is still rendered alongside the fixed-mapping one so the two can be compared directly on the same simulation.

Boundary treatment. A simulator's depth buffer has ideal, hard edges: a silhouette boundary is a one-pixel step. Estimated depth maps, which is what the control branch has seen, do not look like that—their boundaries are soft. Each grayscale frame therefore receives a small symmetric Gaussian blur before encoding. Because the kernel is symmetric, the half-intensity contour of an edge stays where it was, so the smoothing changes how the boundary is presented without displacing the geometry it represents.

Silhouettes, and when they replace depth. The segmentation buffer labels each pixel with the geometry that produced it. Collecting the labels that belong to the simulation's moving bodies and painting them white on black yields a silhouette control: the moving bodies' outlines over time, with no distance information at all. Depth is the default because it carries both outline and distance ordering. It stops being informative when a scene is flat—when every object rests on one near-horizontal surface at nearly the same distance from the camera, the depth field is almost constant across the objects and their surroundings, and the control has little geometric contrast to offer. Measured on the rendered controls, the depth separation between a moving body and the surface immediately around it is about twenty grey levels for a ball bouncing on a floor and about a hundred for a ball arcing through open air, but only about fourteen for a rack of balls on a tabletop. For those flat layouts we substitute the silhouette, which restores a maximum-contrast figure-and-ground signal; the price is that distance ordering between the objects is discarded, and the generator has to recover it from the prompt and from ordinary scene priors.

A timing property of the renders. The frame budget and playback rate are fixed—eighty-one frames written at sixteen frames per second—while the simulated interval differs per scene. The clip therefore plays the simulated motion back stretched in time, by factors between roughly 1.8× and 2.3× for the scenes used here. The trajectory shapes and the contact events are the simulation's; their absolute rate in the finished clip is not, and a downstream measurement of speed or duration must be read against the render clock rather than the simulator's.

Relation to structure conditioning generally. Driving synthesis from an explicit geometric control rather than from text alone follows the lineage of ControlNet for images [57] and motion and structure control for video [25, 49]. What differs here is the provenance of the control: it is not estimated from a photograph or drawn by hand but computed from a physics solver, so the structure the generator is asked to follow is physically consistent by construction.

Setting	**Value**
Backbone	Wan 2.2-VACE-Fun-14B (two-expert MoE), diffusers `WanVACEPipeline`, bf16
Memory	experts CPU-offloaded (two-expert weights exceed an 80 GB accelerator)
Control signal	MuJoCo z-buffer depth (globally normalized); object silhouette for flat scenes
Conditioning	control video + all-white mask + text prompt; no reference image
Resolution	848×480 (flow-shift 3.0)
Frames / rate	81 frames at 16 fps
Sampler / steps	UniPC, 50 steps
Guidance scale	5.0
Conditioning scale	1.0 (swept over $\{0.5, 0.8, 1.0\}$)

Table 1 Deployed generation configuration, from the generation scripts. A small checkpoint provides a pre-flight sanity gate and is not used for final clips.

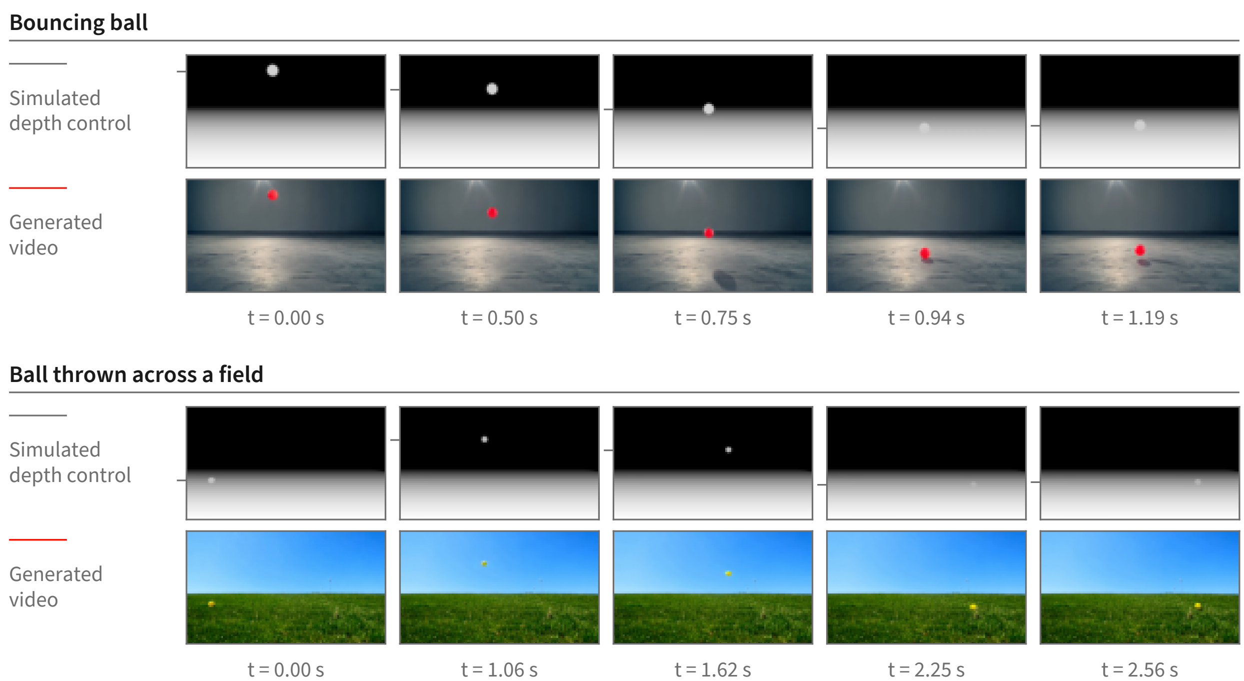


Figure 2 The same mechanism across two scenes with very different requested appearance—an indoor studio and an outdoor field—each shown as its depth control above and the resulting generated clip below. Each scene has its own timestamps, chosen from the turning points of that scene's tracked vertical path so every column carries motion. The appearance changes completely with the prompt while the motion stays the simulation's: for the thrown ball the generated and control paths agree to a correlation of 1.0000 horizontally and 0.9998 vertically, with a median disagreement of 0.7 pixels across and 1.1 pixels down, over the 63 frames in which the ball is measurable in both before it leaves the frame.

2.3 Pipeline and configuration

Table 1 lists the operating configuration. Generation runs as a staged GPU job: control videos are rendered, a small checkpoint provides a fast sanity gate, the 14B checkpoint generates the clips (data-parallel when more than one GPU is available), and the clips are streamed to storage.

2.4 Design choices

Two choices define the deployed recipe. **Native control rather than source re-synthesis:** a control branch consuming a pre-computed depth video keeps the generator from importing the rendered source's synthetic look, which re-synthesizing the simulation frames directly does not avoid. **No reference image:** the pipeline accepts an optional appearance reference, but the deployed recipe omits it, because including it degraded single-object trajectories under inspection.

2.5 Scene library and coverage

The two cases above are single rigid bodies because those are the scenes whose trajectory can be checked against a closed-form path. The scene set itself is much wider: 1,314 scenes across 66 kinds. Before any of them was rendered photorealistically, 652 of the underlying simulations were watched and 591 judged to show the intended physical behaviour clearly enough to be worth generating from; that review is of the simulations, not of the generated video. figure 3 shows twelve of those kinds as real generated frames—cloth draping and crumpling, a flag in wind, a swinging rope and a hanging chain, grains pouring into a heap and draining through a funnel, a ball dropped into a pit of balls, a loaded boat, a cascade down a slope, toppling dominoes and a chaotic two-arm linkage. figure 4 shows three of them the same way figure 2 shows a ball: the simulated depth control above, the generated clip below, at matched timestamps.

One limit should be read off these figures explicitly. Everything here comes out of the same rigid-body engine, so the deformables are that engine's particle-and-link primitives: cloth is a sheet of particles joined by springs, a rope or chain is a short row of linked segments, sand and marbles are many small solid bodies in contact, and floating is a buoyancy force on a solid body in a still fluid volume. None of it is a fluid solve, none of it is smoke, and none of it is a finite-element soft body. The scene captions name the primitive rather than the phenomenon for exactly this reason.

2.6 What the generator achieves

Single-object scenes—a bouncing ball, a projectile arc, a swinging pendulum—render cleanly, with the generated object following the simulated trajectory (figure 2). Multi-object interaction, such as a billiards break, remains open: neither the deployed backbone nor the alternates tried against it produce a clean multi-object collision.

These judgments are qualitative. They rest on visual inspection of the output clips and on a color-blob trajectory tracker that checks whether the generated object follows the control's path. No learned or quantitative video-quality score—FVD, a human-preference model, or a critic score—is applied to this line.

2.7 When the control can be released

The control does not have to be applied for the whole of denoising. A diffusion model builds a video over a sequence of denoising steps that starts from almost pure noise and ends at a clean frame, and the control signal is added in at every one of those steps. That raises a question with both a practical and a scientific side: if the control is switched off partway through, does the motion it has already induced survive to the finished clip? Practically, the steps after the release are free of the control, so the model is free to invent appearance there. Scientifically, the earliest release point that still preserves the motion is the point at which the model has *committed* to that motion.

Method. We pick a step, keep the control at full strength for every step before it, and set the control's contribution to exactly zero from that step onward. Nothing else changes: same simulation, same control video, same prompt, same sampler, same seed. The comparison is always against the same clip generated with the control never released, so the release point is the only difference between the two.

The release point is reported as a noise level, not a step number. A step index is not a property of the model; it is a property of the sampling schedule we happened to choose. The schedule maps each step to a noise level, and that mapping changes when the number of steps or the schedule's shape changes. Concretely, on

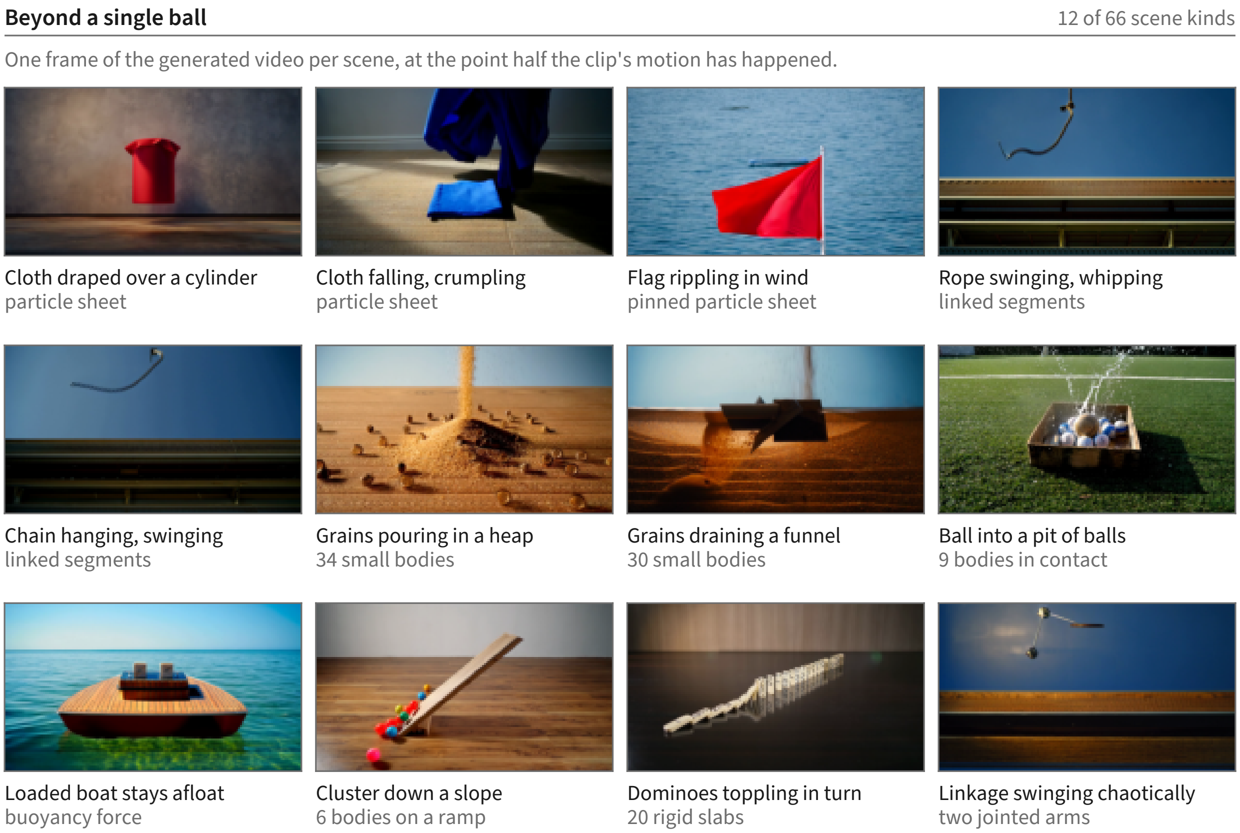


Figure 3 Twelve of the 66 scene kinds, one real generated frame each. The frame shown is computed, not picked: for every clip the frame-to-frame change is accumulated over the whole clip and the tile is the frame by which half of that total change has happened, which lands mid-drape, mid-swing or mid-pour rather than on the first frame. The second caption line names the physics primitive, and every body count in it is the scene's own parameter. The simulation each clip was generated from passed the pre-generation review described in the text; the generated clips themselves were checked by eye for this figure, which is a weaker guarantee than a systematic review. Two candidate tiles were cut after tracking their control renders showed the phenomenon their label would have claimed does not occur in the clip.

the schedule used here, step 8 of 40 sits at a noise level of $\sigma = 0.952$; on a differently-shaped 40-step schedule the same step 8 sits at $\sigma = 0.923$, and the noise level 0.952 has moved to step 5. Reporting "released at step 8" would therefore describe our schedule rather than the model, so every release point below is given as the noise level σ at which the control was switched off, with the step index alongside it only as the label of the run.

What is measured. Both the control video and the generated clip are tracked frame by frame, giving the object's path in each. *Retention* is how much of the simulated motion the generated clip keeps: the agreement between the generated path and the control's path, measured separately across the frame (horizontal) and up and down (vertical), because those two directions do not behave the same way. Agreement is a correlation over the frames in which the object is located in both videos, so a value near 1 means the generated object moved the way the simulation did. Two windows are reported, because a single whole-clip number hides where a failure lives: the *airborne* window, up to the frame at which the simulated object first reaches the ground, and the *after-contact* window from that frame to the end. The contact frame is not hand-set per scene; it is read off the control's own vertical path as the first return to the floor after the object's highest point. Alongside retention we count two failure modes. A *collapse* is a clip whose horizontal agreement falls below 0.7, meaning the sideways motion no longer resembles the simulation's. A *phantom* is the specific disagreement in which the whole-clip number reads as broken (below 0.7) while the airborne window alone reads as clean (above 0.9) — the signature of a clip whose flight is right and whose ending is wrong. Both thresholds were fixed before scoring.

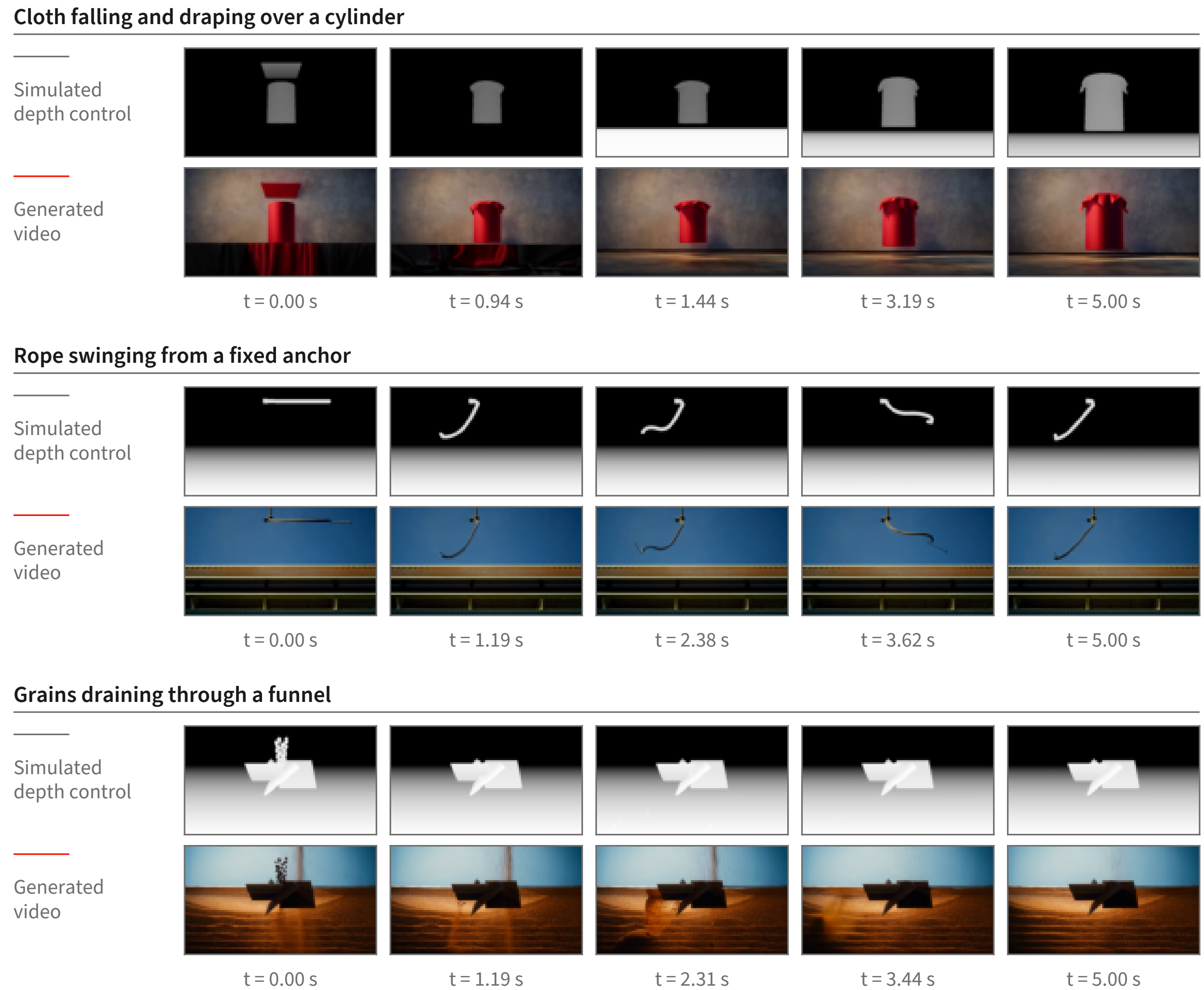


Figure 4 Three deformable and granular scenes shown as figure 2 shows a ball: the simulated depth control above, the generated clip below, at matched timestamps. Columns are placed at equal steps of the clip's accumulated change so no column falls in a static stretch. These three cases are the ones whose depth control still shows its object in every column—in a depth picture an object that settles onto the floor takes on the same grey as the floor behind it and disappears from the control, so how much of each control frame is distinguishable from its background is measured, and cases whose control goes blank are not shown as pairs.

Result: the flight survives an early release; the ground contact does not. Figure 5 shows the measurement. Two scenes were repeated over 16 seeds each at one release point, $\sigma = 0.952$, in two variants (with and without a reset of the sampler's internal history at the release), against the never-released baseline: 96 clips in total, 16 per cell. Figure 5 shows that comparison. Releasing the control at $\sigma = 0.952$ leaves the airborne flight of the thrown ball essentially untouched — per-seed horizontal agreement drops by 0.003 relative to never releasing (95% confidence interval 0.001 to 0.004, $n = 16$ paired seeds), which is a real but negligible difference. Over the whole clip the same release costs 0.13 horizontally (95% CI -0.02 to 0.28, $n = 16$) and 0.18 vertically (95% CI 0.07 to 0.29, $n = 16$). The gap between those two windows is the finding: what an early release breaks is not the arc but what happens after the ball reaches the ground. Scoring only the frames after the simulated contact, the generated ball's horizontal motion runs *opposite* to the simulation's on 9 of 16 seeds when the control is released at $\sigma = 0.952$, and on 0 of 16 when it is never released — the simulation rolls the ball forward and the generator, cut loose, rolls it back. The whole-clip collapse count follows: 3 of 16 seeds at $\sigma = 0.952$ versus 1 of 16 never released, and the phantom count is 2 of 16 versus 0 of 16, so the collapses that do appear are predominantly ending failures on clips whose flight was clean.

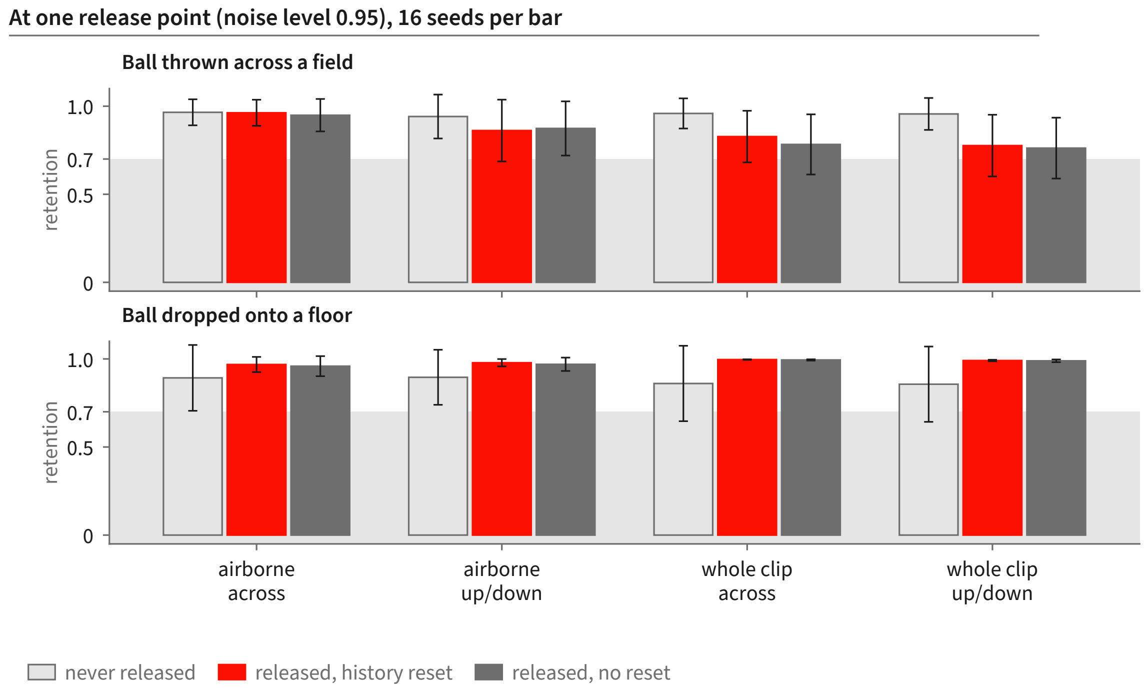


Figure 5 The seeded measurement at the single release point $\sigma = 0.952$, 16 seeds per bar, error bars showing 95% confidence intervals of the mean. Retention is the agreement between the generated object's tracked path and the simulation's, so 1.0 is "moved the way the simulation did". The three bars in each group are the never-released baseline and the two release variants, which differ only in whether the sampler's internal history is also reset at the release. The airborne window is indistinguishable from never releasing; the whole-clip number is not, and the direction that suffers is the horizontal one. Grey band marks the region below the pre-registered 0.7 break line.

Result: a later release is clean, and the vertical direction is the robust one. Sweeping the release point across six settings on a single seed for three scenes gives the shape of the curve (figure 6). For the thrown ball, releasing at $\sigma = 0.952$ gives an after-contact horizontal agreement of -0.63 (the ball travels the wrong way) with the airborne window still at 1.00; releasing at $\sigma = 0.903$ or later restores the after-contact window to 0.98 or above, and it stays there through the latest release point tested, $\sigma = 0.555$. The same sweep on a ball dropped onto a floor never breaks at all — after-contact horizontal agreement is 1.00 at every release point including the earliest, and the vertical agreement, which is the informative direction for a straight-down drop, is lowest at the earliest release (0.78 at $\sigma = 0.952$) and rises to 0.99 or above once the release is at $\sigma = 0.903$ or later. A ball rolling off a ramp holds 0.99 or better at every release point in both directions, but only a third of its frames are measurable at all, so it constrains little. Across all three scenes, vertical motion — the direction gravity supplies — is markedly more robust to an early release than horizontal motion, which the simulation alone specifies.

A denser sweep, and what it does not settle. A third run sweeps three release points on the thrown ball at three seeds and two guidance strengths, 18 clips, and scores them against the control with a segmentation-based tracker rather than a colour tracker. It reproduces the direction of the effect — airborne horizontal agreement is 1.00 (mean over the 6 clips at $\sigma = 0.952$, 95% CI 0.998 to 1.000), while after-contact horizontal agreement at that same release point averages 0.69 with a confidence interval spanning zero (-0.04 to 1.43, $n = 6$) and is negative on one clip; at $\sigma = 0.903$ and $\sigma = 0.833$ the after-contact window recovers to 0.99 and 1.00. But it also shows the ceiling on this measurement: at the latest release point the tracker loses the object entirely on 3 of 6 clips, so those cells rest on 3 clips, not 6. On inspection those are clips in which the generated ball is genuinely tiny and low-contrast against the grass, not clips in which the tracker mis-fired, but either way the confidence intervals at the late release points are too wide to place the transition sharply.

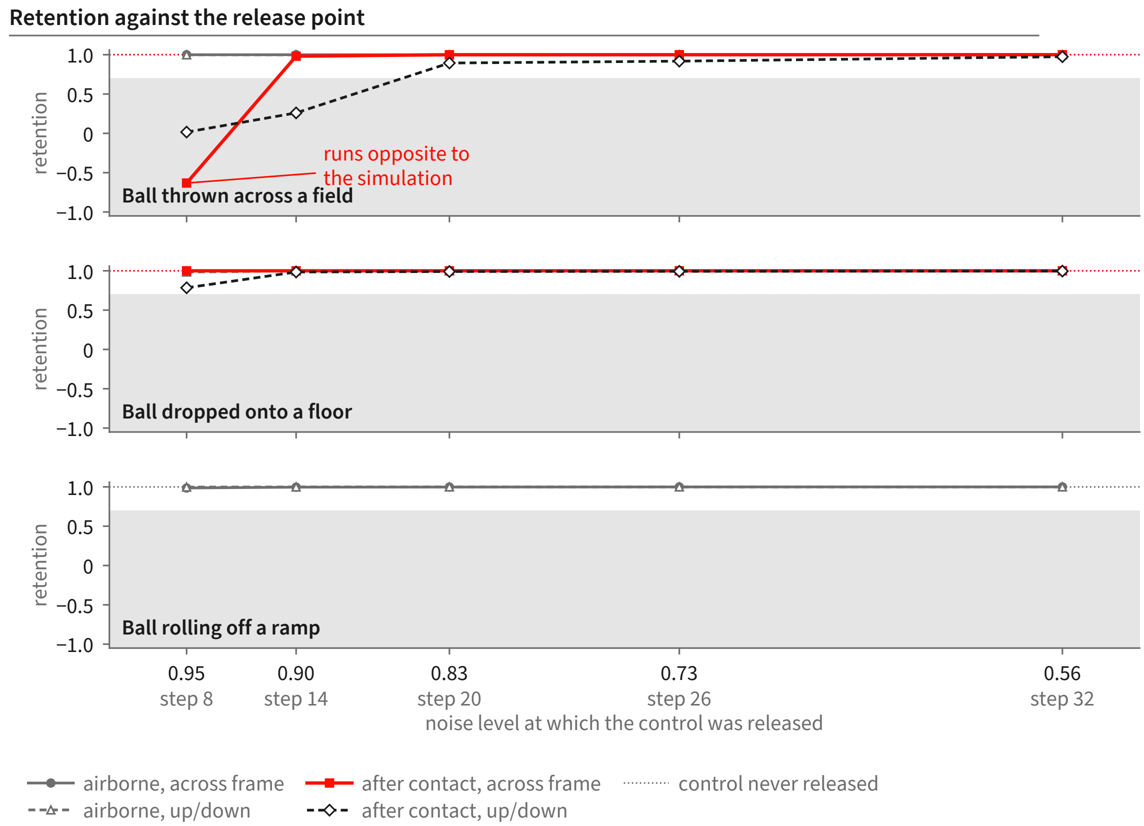


Figure 6 The release-point sweep: one clip per point, three scenes, scored separately over the airborne window and the window after the simulated object first reaches the ground. The horizontal axis is the noise level at which the control was switched off, high noise on the left; the step index of each run is printed underneath as a label only, because the same step index sits at a different noise level under a different schedule. The thrown ball's after-contact horizontal motion is the one curve that breaks, and it breaks only at the earliest release: it runs *opposite* to the simulation at $\sigma = 0.952$ and is restored from $\sigma = 0.903$ onward. Its airborne flight is unaffected at every release point. Across all three scenes, vertical motion — the direction gravity supplies — is markedly more robust to an early release than horizontal motion, which the simulation alone specifies. Grey band marks the region below the pre-registered 0.7 break line.

Scope. These numbers characterize one conditioning route. The runs condition the backbone by carrying a rendered source video through the denoising process and releasing it partway, rather than through the dedicated control branch of section 2. The question and the noise schedule are the same in both cases, but the release point measured here is a property of the source-carrying route, and its transfer to the control branch is untested.

The scenes are single moving rigid bodies—a ball thrown across a field, a ball dropped onto a floor, a ball rolling off a ramp—at one resolution and frame count. Deformable and granular scenes (section 2.5) are outside the study, because the tracker cannot follow their objects as a single path. Statistical depth is uneven: one release point carries 16 seeds, while the shape of the curve across release points rests on one seed per point for three scenes and three seeds per point for a fourth.

One factor is not isolated. In one of the two variants the release is accompanied by a sampler-history reset, and the study does not separate the two. The variants are statistically indistinguishable on every window and direction measured—the largest per-seed difference is 0.04 horizontally over the whole clip, 95% CI -0.04 to 0.13, $n = 16$—which is what one expects if the reset does nothing once the motion has committed, but it does not establish that it never matters.

Field	What it holds	Coverage
violated fragment	the words of the prompt the video fails, quoted verbatim; median 8 words	306/306
rationale	free text saying what went wrong instead; median 25 words	306/306
severity	1–5; the distribution is centred at 3 and 4	306/306
confidence	1–5; 239 of 306 are marked 4 or 5	306/306
time span	first and last frame in which the failure is visible	237/306
box track	boxes at the start, middle and end of that span, one per object involved, in normalized coordinates	208/306

Table 2 The six fields of a flaw record and how often each is filled across the 150-clip core. The first four are always present; timing and boxes are not, and a flaw carrying neither is one whose complaint is about the clip as a whole or about something that never appears.

3 A Human-Annotated Video-Flaw Benchmark

The benchmark is a corpus of 1,500 AI-generated clips in which a person watched each clip against its prompt and wrote down every place the video failed to deliver what the text asked for. Every clip carries at least one such record. The generating model is not recorded in the data, and no clip is attributed to a specific system. The human-written fields are the ground truth throughout this report.

3.1 What a flaw record contains

A flaw is a localized record with six fields. Four are always present: the **violated fragment**, the words of the prompt the video fails, quoted verbatim; a free-text **rationale**; a **severity**; and the annotator's **con[]dence**. Two are present when the annotator could supply them: a **time span**, the first and last frame in which the failure is visible, and a **box track**, boxes at the start, middle and end of that span, one per object involved, in normalized coordinates. Table 2 gives each field's measured coverage across the evaluation core, and figure 7 shows one record in full. Each clip additionally carries an overall severity and a short list of the objects, actions and relations its prompt calls for.

Two coarse tags support grouping and reporting: a *modality* tag saying whether the complaint concerns what is seen, what is heard, or both, and a *type* tag naming the dominant kind of complaint from a fixed ten-item list. Neither is written by the annotator. Both are assigned by three language models classifying each flaw from its quoted fragment and rationale at temperature zero under a fixed seed, resolved by majority; a three-way disagreement resolves to "both" for modality and "other" for type, and a flaw receiving no usable vote falls back to a frozen keyword rule. They are labelled machine-assigned wherever they appear and are never treated as ground truth. Annotation is single-annotator: the corpus carries one annotation per flaw, so no inter-annotator agreement figure is available.

3.2 How the evaluation set is derived

Selection runs in three recorded stages, and the denominator of every later measurement depends on them. Table 3 gives the counts and figure 8 the same derivation as a sequence of steps.

Stage one, clip selection. A clip enters the pool if its video is retrievable and it shows signs of dynamics. The dynamics test is a fixed vocabulary of motion words—falling, bouncing, rolling, colliding, pouring, toppling—applied to the prompt and, separately, to each flaw's rationale and quoted fragment; a clip is kept when its prompt contains one, or one of its flaws does, or it has a time-windowed flaw against an action-typed object. The test is a keyword rule and selects for prompts and complaints that *talk about* motion, which is a proxy for, not a guarantee of, a flaw whose adjudication needs physical reasoning. This stage keeps 606 of the 1,500 clips, carrying 1,171 flaws.

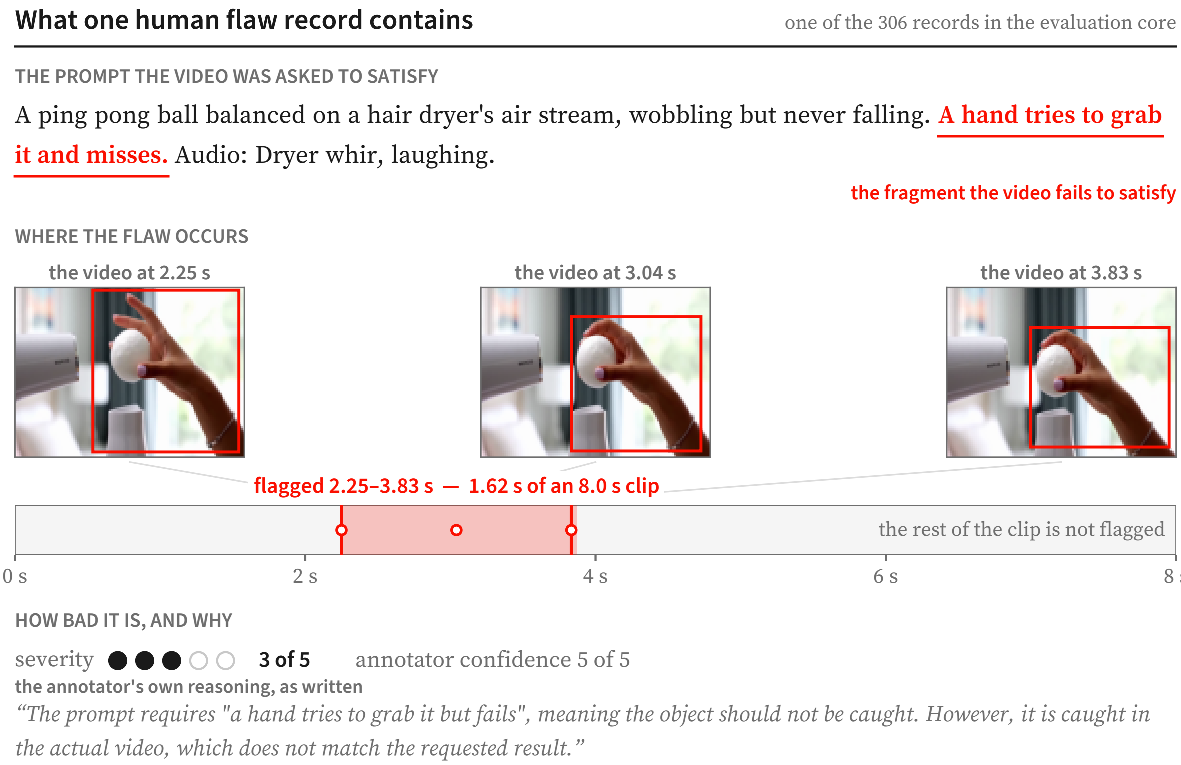


Figure 7 One human flaw record, shown in full. The annotator quotes the prompt fragment the video fails to satisfy, bounds the interval in which the failure is visible, boxes the objects involved at the start, middle and end of that interval, and writes a severity, a confidence and a rationale in their own words. Frames and boxes are the annotator's; the flagged window is 1.62 s of an 8 s clip. Nothing about the critic's output appears here.

Stage two, flaw selection. Within those clips, a flaw is a usable detection target only if a critic seeing only pixels could in principle find it. Flaws tagged "seen" or "both" are kept and purely audible complaints dropped, which removes 64 flaws and leaves 1,107 as the full evaluation set.

Stage three, the frozen core. A core of 150 clips with 306 flaws is drawn for routine measurement. The draw is deterministic and its contract recorded: a fixed seed; strata crossing overall clip severity, flaw count per clip and clip duration; largest-remainder proportional allocation across strata; within each stratum, candidates ordered lexicographically then shuffled once by a generator whose call sequence is fixed; and a replacement rule drawing from the same stratum for any clip that cannot be downloaded or decoded. A continuity-focused stratum of 29 clips and 85 flaws sits inside the core and is exempt from replacement. Every clip is downloaded and decoded before the core is written, and the selection, strata and replacements are stored with the set.

The core is an *optimization set*: system and prompt decisions are made against it, so results on it measure fit to that core, and a generalization claim requires a set that has not influenced system design. A separate pool of 711 annotated clips is held disjoint from both frozen sets for any learning that consumes this data.

3.3 What is in the evaluation core

The core is 150 clips, median 8.0 s long (range 5.0–10.3 s), mostly at 24 fps (118 of 150; the rest at 16, 25, 30 or 32). The prompts are not one-line captions: a median prompt is 42 words and the longest 188, and many specify a shot list, a count, an ordering, an on-screen string and a sound in one paragraph. Sixty-six of the 150 clips carry flaws of more than one kind, so a single number per clip cannot say which of them a detector found.

Its 306 flaws, a median of two per clip, group by machine-assigned type as: actions that do not happen or

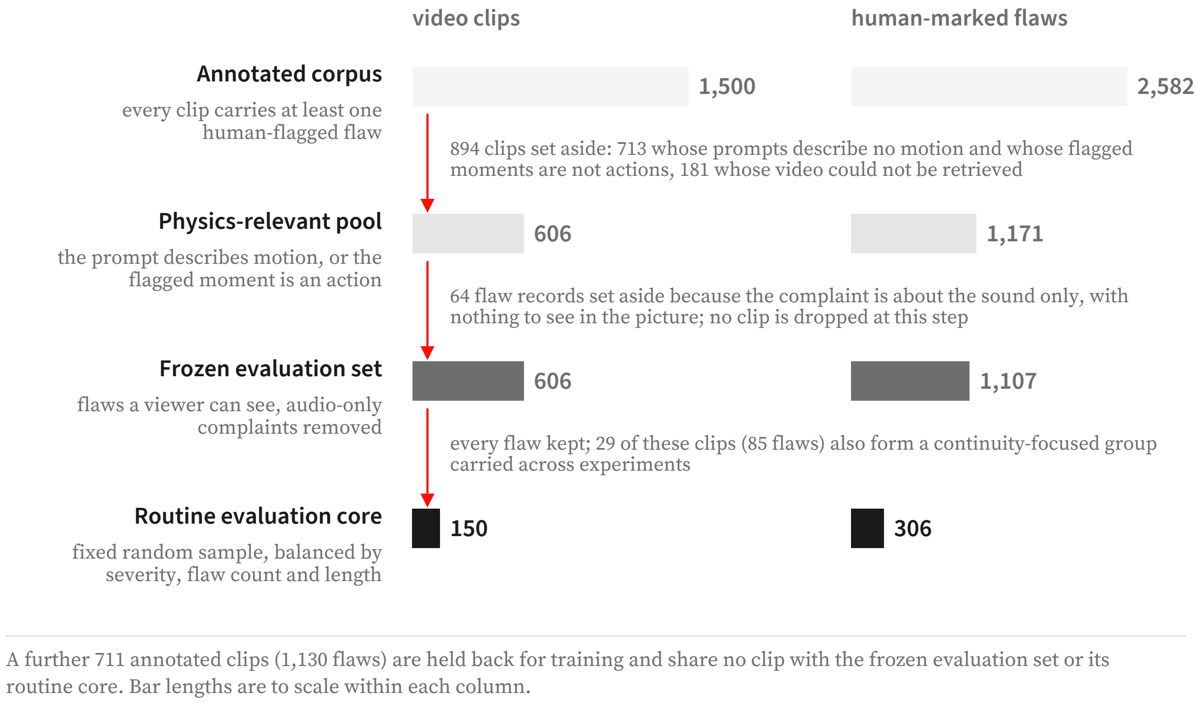


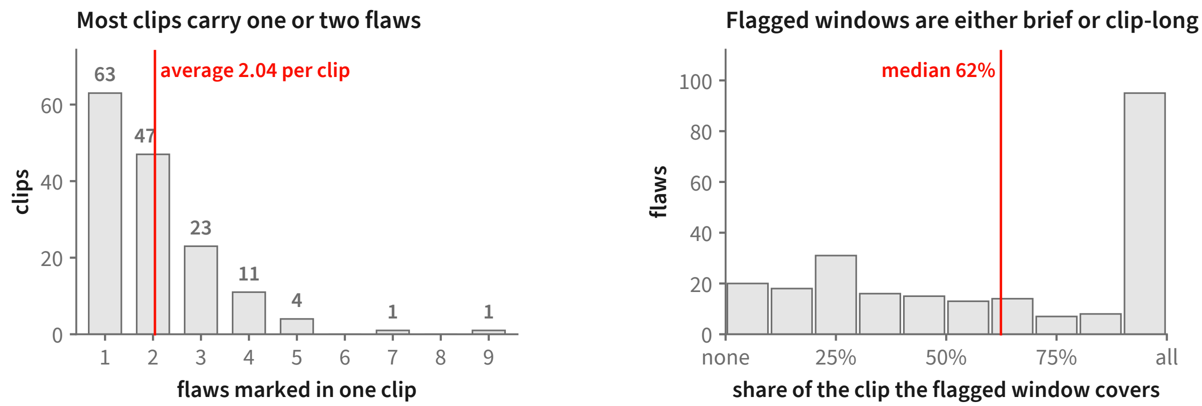
Figure 8 How the evaluation set is derived from the annotated corpus. Clips are removed only at the first step; the second step removes flaw *records* whose complaint is audible but not visible, leaving the clip count unchanged. The routine core is a fixed random sample balanced by severity, flaw count and clip length. A further 711 annotated clips are kept aside for training and share no clip with either frozen set.

Set	Clips	Human flaws
Full corpus (all AI-generated, all annotated)	1,500	2,582
Physics-relevant pool	606	1,171
Full evaluation set (frozen)	606	1,107
Routine evaluation core (frozen)	150	306
continuity stratum within the core	29	85
Disjoint training pool (excludes the frozen sets)	711	—

Table 3 Benchmark statistics, from the dataset manifests. The full evaluation set drops a small number of audio-scoped flaws relative to the physics-relevant pool. The training pool excludes every clip in the frozen evaluation sets.

happen wrongly, 139; wrong counts, 36; wrong or missing objects, 30; garbled on-screen text, 25; wrong spatial arrangement, 19; wrong ordering in time, 17; wrong attributes such as colour or material, 16; camera and style, 10; physical motion, 7; and a residual 7. The largest category is whether the requested action occurred at all, so this is not a physics benchmark in the narrow sense. Counting, text and spatial arrangement together account for 80 flaws, each a question a general-purpose model answers poorly and a dedicated measurement answers well.

Of the 237 flaws carrying a time span, the flagged window covers a median of 62% of the clip, and the distribution is bimodal: 57 are confined to a quarter of the clip or less, and 95 run for at least nine-tenths of it (figure 9). A detector needs both a way to look closely at a moment and a way to judge a whole clip.



150 clips of the routine evaluation core, 306 human-marked flaws. The right-hand panel uses the 237 flaws that carry a start and an end frame; the remaining 69 were marked without one. Of those 237, 57 fit inside a quarter of their clip and 95 run almost its whole length; the median flagged window lasts 5.00 s.

Figure 9 Two properties of the routine evaluation core that shape how a detector must behave. **Left:** most clips are flagged in only one or two places, so a detector that raises many complaints per clip will mostly be raising ones no annotator marked. **Right:** flagged windows are bimodal — many failures are confined to a small part of the clip while a comparable group runs almost its whole length — so both a brief localized check and a whole-clip check are needed.

3.4 Scoring conventions

Every clip carries at least one flaw. With no clean clips, recall is directly measurable and precision is not: a detector that flags everything scores perfectly on recall alone, so results on this set are reported alongside the rate at which accusations land on a real flaw.

Scoring is over *distinct* flaw spans. Twelve clips record the same failure two or three times under different identifiers—eighteen duplicate entries across the core—and an accusation can be credited to at most one flaw, so identical spans on a clip count as one flaw. The core's 306 entries are therefore 288 distinct flaws. A run that covers a subset of clips carries its own share of both counts; section 7 states them.

3.5 Relation to existing video benchmarks

The closest relative is VideoPhy-2 [6], where three annotators rate each clip on two five-point scales—semantic adherence and physical commonsense—and mark each candidate physical rule as violated, followed or undetermined, with scores averaged and rules resolved by majority. The judgement attaches to the clip and to a predefined rule; annotators do not mark frames, spans or regions. Benchmarks built around per-dimension automatic metrics, such as VBench [23] and T2V-CompBench [45], score a model over their own fixed prompt suite, which makes them instruments for ranking generators rather than for scoring a detector on someone else's clips.

Here the judgement attaches to the words of the prompt that failed, and, where present, to a time span and a box track. That is what makes a detector's output matchable rather than merely correlatable, and it is what supports the miss attribution of section 7. A clip-level score offers nothing to align an individual accusation against.

The trade runs the other way as well. VideoPhy-2's three annotators with majority resolution give a redundancy this benchmark lacks. A user who needs a label whose noise is characterised should prefer the redundant benchmark; a user who needs to know which failure a detector found, and where, needs this one.

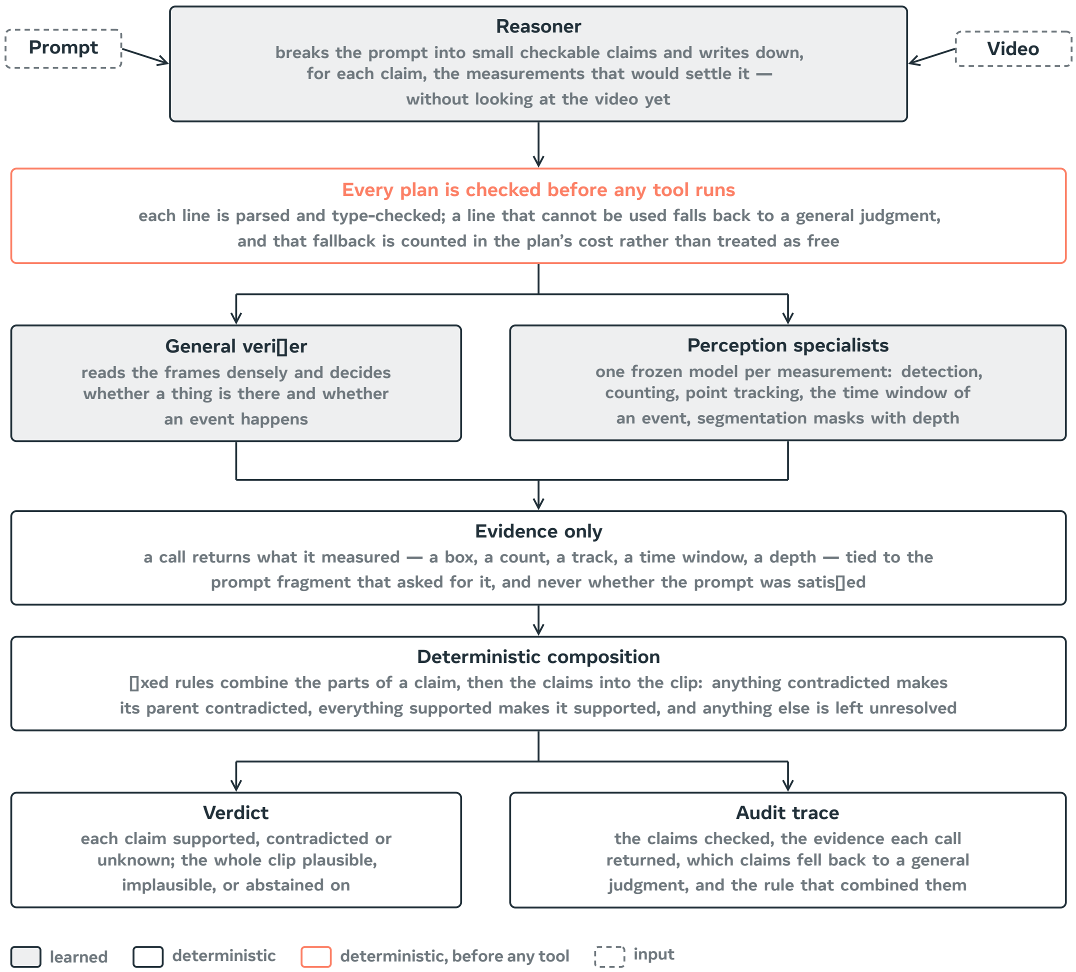


Figure 10 The critic. A reasoner turns the prompt into checkable claims and writes down the measurements that would settle each one; the plan is type-checked before any tool runs; a general verifier reading the frames densely decides whether things are present and events happen, while frozen specialists supply the quantities it cannot produce. Every call returns evidence only, and fixed rules — not a second model — turn that evidence into the verdict and the trace beside it. Shaded boxes are learned components; outlined boxes are deterministic.

4 The Auditable Multi-Tool Critic

4.1 The reasoner

One served vision-language model—Qwen3-VL-30B-A3B-Instruct [39], BF16 weights (no quantization), tensor-parallel over two GPUs—performs the decomposition and grounding and runs three reasoner-mediated hard sub-checks (section 4.5). It is the only learned component that reasons over the whole prompt. Figure 10 shows how it sits between the prompt and the specialists that supply the evidence.

4.2 Perception specialists

Each claim is typed, and a claim whose type names a measurable quantity—a count, an interval, an outline, a depth ordering, a track, a transcription, a sound—is routed to a frozen open specialist that measures that one thing and returns only the measurement (table 4). Claims that merely assert something occurs are put to the reasoner reading the frames. On the evaluation data those dominate, so most executed checks are settled without a specialist (section 7). Two qualifications apply to the table. First, some adapters *declare* a stronger primary backend that is not the one actually executed; we list the backend that runs. Second, there is no holistic scoring tool in the verdict path by design: a learned holistic video scorer is kept out so that the

Capability	Open backend	Evidence returned
Object presence	LLMDet open-vocabulary detector [13]	presence; bounding box
Object count	CountGD-Box counter [2, 3] (SAM 2.1 internal)	cardinality of a named set
Kinematics	point tracking, TAPNext-style [59] + kinematic predicates	per-frame positions; static/moving, direction, speed
Action timing	the reasoner over timestamped frames (a dedicated temporal grounder [58] remains available on request)	time window of a named event, or a stated refusal
Spatial relation	unified SAM 3 masks [8] + monocular depth + reasoning	depth-ordered / image-plane relations
Rendered text	an optical character recognition engine, over the frames where the surface is on screen	the characters actually printed, compared to the demanded string
Holistic	*no tool by design* [20]	—
Audio†	FlexSED sound-event detection [18]	sound events with timestamps

Table 4 Perception specialists and the open backend wired to each. In the routed configuration the object, count, kinematics, action, and spatial-relation backends execute as stages; the object and count adapters declare stronger primaries that are not on the executed path, so the running backend is shown. The relation capability is served by a unified mask-plus-depth reasoning arm, and no holistic scoring tool is used in the verdict path by design. †The audio detector answers sound claims on the clip's own audio track, and returns the time spans in which it heard the queried sound, so a sound can be ordered against something seen.

verdict is built from localized evidence and deterministic rules, though such a scorer could still serve as an external baseline.

4.3 Evidence-only tool contract

Specialists are frozen, and what they are permitted to say is fixed by a schema rather than by convention. A specialist reports *what it measured*—a box, a count, a track, a time window, a depth—and never *whether the prompt is satisfied*. The schema is enforced by a validator that refuses, rather than repairs, anything off-contract, so an off-contract output cannot enter the accepted evidence record at all. The complete rule set is:

- **Judgment-shaped fields are refused by name.** A fixed list of field names is rejected anywhere in a measurement, at any nesting depth—among them *label*, *judgment*, *decision*, *plausible*, *reasoning*, *rationale*, *ground truth*, *human label*, and *flaw*—as is any field name containing *mismatch* or *verdict*, and any name that starts like an evaluation label.
- **Renaming does not evade the rule.** Field names are normalized before the check: case and separators are stripped and camel-case is split, so a rejected name cannot be smuggled in by rewriting it.
- **Numbers must be real numbers.** Measurements are validated as strict data: no infinities, no not-a-number, no duplicate keys, no values outside the serialisable types. A measurement that cannot be canonically serialised is refused.
- **Every measurement is tied to the text that asked for it.** A record carries the clip identifier, the full prompt, and the exact claim fragment being checked, and that fragment is required to occur *verbatim* in the prompt—a paraphrase is a validation failure. This is what makes a measurement attributable after the fact.
- **Not-measuring is a first-class, and separate, outcome.** A record's status is exactly one of measured, abstained, or errored. A measured record must carry measurements and may carry neither an abstention reason nor an error message; an abstention must carry a reason and no error; an error must carry a message and no abstention reason. The three cannot be blurred.
- **Heavy evidence is content-addressed.** Any file a specialist produces is referenced by path plus a hash of

its bytes, and the measurement payload itself carries a hash of the tool's native output, so a record can be checked later against the artifact it claims to describe.

- **Cost travels with evidence.** Every record carries its own accounting: accelerator seconds and tool calls are required; wall time, price, and token counts are optional. Cost is therefore auditable per measurement rather than reconstructed afterwards.
- **The contract is versioned.** Each record names the schema version it was written against, and a reader refuses a version it does not implement.

The guarantee this buys is narrow and worth stating precisely: no specialist *emits* the verdict, and every measurement that reaches the deterministic layer is attributable to a prompt fragment and to a hashed artifact. It does not make a learned measurement semantically neutral, and it cannot detect a measurement that is confidently wrong; a wrong number inside a well-formed record will mislead the rule that consumes it.

4.4 Three-valued semantics and deterministic roll-up

Every claim resolves to **supported**, **contradicted**, or **unknown**, and **unknown** is first-class: it is what the critic reports when the evidence it actually obtained does not settle the question. The rules that produce those three values are fixed and are stated here in full, because which of them applies is the whole substance of the critic's answer.

A measurement that looked and did not find is evidence, not silence. When a specialist is asked about something over the whole clip and reports that it is not there, that is treated as a measurement and it contradicts the claim; it does not become **unknown**. Concretely: a stated count that does not match the measured count contradicts; a spatial claim whose two participants are never both visible with usable outlines contradicts, because the asserted arrangement was never realised anywhere; an asserted approach or departure whose subject cannot be followed at all contradicts; and a claim that something enters a container is contradicted both when it is never inside and when it is inside from the first observed frame with no crossing ever seen.

Existence and occurrence are decided by the dense-frame model, not by the specialists. A detector that returns a box, or a temporal grounder that returns a window, is not by itself evidence that the queried thing exists: those interfaces return their best candidate for a query whether or not the query has a referent. So whether an object is present at all, and whether a named event happens at all, are routed to the vision-language model watching the frames densely, and the specialists supply the quantities, windows, outlines, depths and tracks it cannot produce. A detector's box remains an *input* to measurements—seeding a track, naming the participants of a spatial relation, feeding a count—and in that role its looked-and-absent report still contradicts. A measurement whose only consumers were existence or occurrence questions is not executed at all, which keeps unused perception out of the cost.

This is not a precaution against a hypothetical failure. Asked to time eleven events that a human viewer had confirmed do *not* occur in their clips, a dedicated temporal grounding model returned a confident time window for all eleven—placing tigers crouching across seven seconds of an eight-second clip, and a candle going out in a clip where the flame burns to the end—while the same reasoner that watches the frames, asked the same eleven questions in a form that permits refusal, said the event was absent in seven of them. On a control set of six events that do occur in the same clips, the reasoner found five and the grounder six, so the difference is not a reasoner that refuses everything. Where a window *was* readable the grounder ordered events perfectly; its failures were unusable output presented as measurement. Consequently the event-timing question is now put to the reasoner over frames labelled with their timestamps, in a form whose permitted answers include *absent* and *present but not placeable in time*; the dedicated grounder remains available when a caller asks for it. Two windows that begin at the same instant, or that each span the whole clip, are marked as carrying no order and the ordered claim is left unresolved rather than decided by a tie.

The comparison is small—eleven absent events and six present ones, on clips a single viewer had annotated—so it fixes the routing of one question rather than ranking two models in general, and it says nothing about how often either backend is right when an event *is* present and placeable. What it does establish is the asymmetry the design turns on: a backend that cannot decline will answer an unanswerable question, and an invented window is indistinguishable from a measured one downstream.

Permitting refusal is not free, and one traced case shows both the cost and the reason to pay it. A claim asserted that a sound coincided with each impact. Asked to place that sound in time, the reasoner declined, stating that the frames carry no audio and that the question could not be answered from them. The ordering was left unresolved, and a catch was lost: the previous backend had returned a window for the same query without any audio evidence, and that unsupported estimate produced a flag matching a human-labelled mismatch.

A reader that cannot be fluent. The same asymmetry decides who reads text on screen. Generated video often renders lettering that is shaped like writing but is not writing, and that is precisely the defect a claim quoting a sign or a slogan is asking about. A generative model reading those pixels supplies the missing letters from its language prior: shown a crop of one camera body, the vision-language model returned *"sound and colour, for film and sound recording"*, and—once told what the prompt had asked for—*"shoot the camera to the front and enjoy the moment"*. Neither sentence is on the camera. A recognition engine given the same crop returned *"RRVLE"*, *"ToeFan"*, *"ao Gony Raa um"*, which is what is actually printed there. On two clips where a human annotator had typed out the garbled lettering by hand, the engine's transcript matched theirs character for character. So the reading is done by an engine that transcribes glyphs and cannot compose a plausible sentence, and the comparison against the demanded string is ordinary text comparison. The distinction is not stylistic: a fluent reader would have reported the prompt's own words back and the claim would have been marked satisfied.

Silence from that engine is then split by cause rather than treated as one outcome. If the engine itself fails, the claim is unresolved—that is a statement about the instrument. If the surface was on screen, and large and sharp enough that legible lettering would have survived, and nothing legible was read, the claim is contradicted: illegible lettering is the defect, not an absence of evidence about it. If the pictures were too small or too smeared for any text to survive, the claim is unresolved, which is the only remaining genuine abstention.

The refusal was also correct in a way the invented window concealed. The claim was about sound, and it was being put to a model that sees only frames—so no answer from that backend, right or wrong, could have been evidence. The refusal surfaced a routing fault rather than a perception limit, and sound claims are now measured on the clip's audio track by a detector queried with the described sound (table 5); most clips in the evaluation set carry one. That is the general argument for permitting refusal: a backend that must answer converts a question it cannot address into a confident measurement, and downstream nothing distinguishes that from a real one. The cost is concrete—one catch in this trace—and its aggregate size is not established here.

Ordering is composed, never assumed. For a claim that one event precedes or falls inside another, both events must first be confirmed to occur by a backend able to observe them—the dense-frame model for anything visible, the sound-event detector for anything audible—so a sound can be ordered against something seen. An event no available backend can observe yields a refusal rather than a window. If either event is judged absent, the ordered claim is contradicted outright; if either cannot be placed in time, the claim is left unresolved. Only once both are confirmed do the specialist's time windows decide the order, and they do so with a tolerance rather than a fixed threshold: the boundary slack is a quarter of the shorter of the two events, so "before" requires a gap that is resolvable relative to the events being ordered. Containment is directional—one interval inside the other, not merely overlapping it—and an interval that contains the other contradicts rather than abstains.

Where **unknown** *genuinely comes from.* It is reserved, and it is reserved for cases that are about the evidence rather than about the video. Two valid window measurements that cannot be reconciled within the tolerance yield **unknown**: that is the deterministic outcome of the ordering table, not a tool's opinion. So does an incomplete plan—a depth-ordering relation asked for without a depth measurement is reported as a gap rather than silently resolved from outlines alone. So does a required measurement that never arrived. So does a check that the planner emitted but bound to no claim text and no describable request, which is surfaced by name rather than absorbed. And so does an infrastructure failure, which is retried at the clip level and never scored.

Sub-checks compose by a fixed rule, and an unresolved part does not disappear. A claim may carry several sub-checks. The claim is **contradicted** if *any* of them is contradicted; it is **supported** only if *every* one is supported; in every other case—which is exactly the case where at least one part is unresolved and none is contradicted—the claim itself is left unresolved. An unresolved part therefore blocks a claim from being asserted as satisfied, but does not by itself accuse.

Negation is handled at composition, not inside measurement. A negated sub-check is written only around a whole assertion—"this does not happen", "this object is not present"—and never around a measurement. At composition time its outcome is flipped first, supported becoming contradicted and contradicted becoming supported, and the ordinary rule above then applies. The consequence is that the measurement rules never have to reason about a negated query, which is why negation around a count, an interval or a spatial relation is rejected at planning time rather than approximated.

The clip answer. The roll-up is a separate deterministic step over the per-claim outcomes: if any claim is **contradicted**, the clip is *implausible*; otherwise, if every claim is **supported**, it is *plausible*; otherwise the critic **abstain**s. "Plausible" therefore means only that no checked claim was contradicted—a summary of the plan that was executed, not a certificate of complete physical verification.

Where an abstain-by-default rule does apply. The three reasoner-mediated sub-checks of section 4.5 are the exception, and they are asymmetric in the opposite direction from the measurement rules. Each may only report a mismatch on the strength of a positively contradicting feature it can actually see; if the features that would distinguish the two candidate readings are not clearly resolvable in the evidence it was given, it abstains, and an unparseable or missing answer also abstains. Absence of confirmation is never treated as a contradiction there.

4.5 Reasoner-mediated hard sub-checks

Three checks resist full symbolization and remain reasoner-mediated on tool-localized inputs: object subtype identity (is the detected object the specific kind named), dense action-event recall (did a named event occur, densely over time), and depth-ordered spatial relations. Each runs a fresh reasoner pass over evidence a tool has already localized, under the same asymmetric gate. We name them explicitly rather than imply a pipeline that is more symbolic than it is.

5 Compiling Claims into Executable Measurement Plans

Routing a claim to a tool could be a flat lookup table. Instead, each prompt is compiled into a *verification plan*: a small, typed program that names the measurements its claims require and how their results combine. The plan is written before any frame is read, validated before any tool runs, and executed so that equivalent measurements are computed once and shared. Because the plan is an explicit artifact, the orchestration itself can be inspected, replayed and criticized—which of the two events was timed, which frames a measurement was scoped to, and which rule turned the evidence into a verdict.

The payoff is not bookkeeping. A plan lets one claim's result gate another's, lets a measured interval narrow where a later measurement looks, and lets a check decline to answer when its inputs cannot separate the alternatives (section 5.5). Section 7 measures what this buys against verifying each claim on its own.

5.1 A restricted language for measurement plans

Each claim is represented by **one line of code** in a small, typed language—the code-as-plan idea used in visual programming systems [17, 46], here narrowed to a verification vocabulary. For claim c_N the planner writes `check(cN, <expr>)`, where the expression is a single call or several joined by `and`. The reasoner does not see the

Operation	Meaning	Becomes
`judge("assertion")`	a dense-frame pass over the clip decides the assertion	event occurrence
`exists("object")`	the object is visible at some point	existence
`eq(count("phrase"), N)`	exactly N are visible	count
`before(window(A), window(B))`	A ends before B	temporal order
`during(window(A), window(B))`	A falls within B	temporal containment
`rel("A‖B", relation)`	spatial relation between two segmented objects	spatial
`traj("phrase", relation, target)`	tracked motion of an object	trajectory
`sound("description")`	the described sound is asserted	occurrence†
`text("STRING", carrier=...)`	the clip renders that exact string on the named surface	rendered text

Table 5 The planner's measurement vocabulary. `count` and `window` are structural and legal only inside `eq`, `before`, or `during`. Relations are drawn from fixed vocabularies (for example *behind*, *in front of*, *above*, *contained in* for space; *moves toward*, *rises then falls*, *accelerates* for motion), and the three directed motion relations require an explicit target. Depth-ordering relations (*behind*, *in front of*) cannot be settled from masks alone, so they additionally require a depth measurement; a plan that omits it is rejected as incomplete rather than silently resolved. †A sound claim is a real measurement on the clip's audio track, made by a detector queried with the described sound. If the clip carries no audio track at all, the claim resolves to *unknown* rather than contradicted: a file that was never listened to cannot supply evidence of absence.

video at this stage; it is writing down *what would have to be measured*, not answering. Representing the plan this way avoids cross-referenced identifiers spread across several generated tables and allows the plan to be checked mechanically before execution—planner-format fragility being a reported difficulty in tool-controller systems [43]. Table 5 lists the callable operations and the check each becomes.

5.2 How a plan is constructed

Whatever surface the planner writes, the plan the executor consumes is a small set of linked tables: the things the prompt talks about, the separate occurrences of the events it asserts, the measurements to be made, the typed checks over those measurements, and the claim each check answers. When the planner writes one line per claim, the deterministic translator assembles those tables as it goes, giving one identifier to each distinct mention it encounters and reusing it thereafter. When the tables are written directly, that consistency has to be arranged explicitly, and the construction proceeds in two passes over the prompt (figure 11). The reason is reference, not length.

Name the things once, then decide what to measure. The first pass reads the whole prompt and does nothing but name. It lists every distinct object or actor the claims mention and every separate occurrence of an asserted event, fixes an identifier for each, and records which participants each occurrence involves. Mentions the prompt itself distinguishes—a first and a second animal of the same kind—receive different identifiers, and an event asserted twice receives one entry per occurrence rather than one entry for the event type. The pass is required to return nothing else: no measurements and no checks. Only then does a second pass decide, claim by claim, what would have to be measured, and it does so against that finished list, which it may refer to but not extend.

The split exists to keep references stable. If each part of a long prompt chose its own names while also choosing its own measurements, two parts would name the same object differently, or give one name to two different objects, and a check written in one part could not be composed with a check written in another—an ordering check, for instance, is only meaningful if both events it orders are the events the rest of the plan is talking about. Fixing the names first makes them shared vocabulary: the second pass points at a name and cannot quietly mint a competing one for something already named.

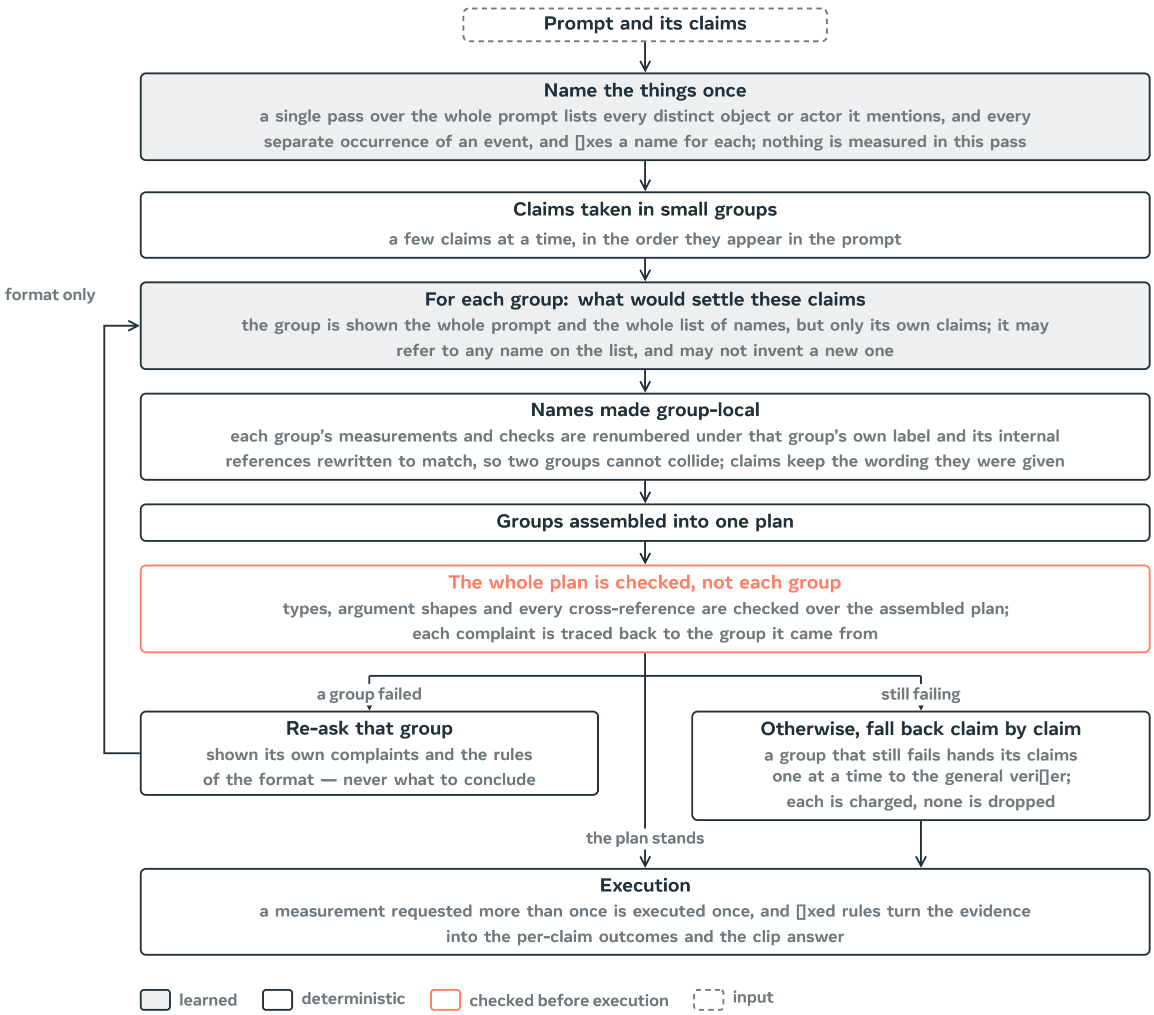


Figure 11 How a plan is constructed. One pass names everything the prompt talks about—the objects and actors, and each separate occurrence of an asserted event—and nothing else; the claims are then handled a few at a time against that fixed list, so that different parts of a long prompt refer to the same things. Each group's identifiers are renumbered under its own label so groups cannot collide when assembled, and the assembled plan is checked as a whole, because what needs checking is mostly cross-reference. A group that fails is asked again with its own complaints and a restatement of the format, never with the answer; a group that still fails hands its claims to the general verifier one at a time, charged to the plan's cost. Shaded boxes are learned components; outlined boxes are deterministic.

Claims are handled in small groups. The claims are then taken a few at a time—four by default—in the order they appear in the prompt. A group is shown the whole prompt and the whole list of names, but only its own claims, and returns only the measurements, the checks and the claim-to-check bindings for those claims. Because groups are written independently, their internal identifiers would collide once the groups are put together, so the identifiers are made group-local mechanically rather than by request: every measurement and check a group produced is renumbered under that group's own label and the references inside the group are rewritten to match. A collision is therefore impossible even if a group ignores the instruction to label its own output. Two things are deliberately not the planner's to choose: the claim identifiers and the claim wording are the ones the claim list already fixed, and a group that leaves one of its claims without a check yields a complaint naming that claim rather than a plan quietly missing it.

The assembled plan is checked as a whole, not group by group. Validation runs once, on the assembled plan. This is not a matter of convenience. Most of what needs checking is a reference from one part of the plan to another: whether a check points at a measurement that exists, whether a claim points at a check that exists, whether an event occurrence is bound to a participant that was actually named. A group inspected on its own cannot be checked for any of it, so group-by-group checking would accept every group and still admit an incoherent plan. Alongside those references the whole-plan check enforces the same typing described in section 5.1: each kind of check must be fed measurements of the kinds that can settle it, relations must come from the fixed vocabularies, a stated quantity must be present, and a directed-motion check must carry a target. Each complaint the check produces is then traced back to the group whose identifiers it mentions.

Recovery is bounded, and it never supplies the answer. A group whose output drew a complaint is asked again, and is shown three things: its own previous output, the complaints against it, and a restatement of the format—the permitted measurement operations, the permitted kinds of check, the labelling convention for identifiers, and the requirement to cover every claim it was given. It is not told which measurement the claim should get, which relation is the right one, or what to conclude. That line matters: the choice of what to measure is the part of the system under study, so correcting it automatically would replace the behaviour being measured with the corrector's behaviour and would hide the planning failures that should be counted. Teaching the format is not the same as supplying the decision, and only the former is done.

Recovery is bounded in both directions. The naming pass and each claim group are allowed a small fixed number of attempts. A group still failing after them is not dropped and is not left unresolved: its claims are handed to the general verifier one at a time, and each of those verifications is charged to the plan's cost exactly like any other call. If the naming pass itself never produces a usable list, there is no shared vocabulary to write measurements against, and every claim of that prompt takes the same per-claim route. After any such degradation the remaining plan is rebuilt and re-checked, and if it still fails, its claims degrade the same way rather than an unchecked plan being passed on to execution.

What this buys and what it costs. What it buys is structural: names are agreed before anything is measured, so checks written for different parts of a long prompt refer to the same objects and events and can be composed; a failure is localised to a group and is either repaired or charged rather than silently shrinking the plan; and no plan reaches execution without passing a check over its complete set of cross-references. What it costs is also concrete. Construction takes several model calls rather than one—one for naming, one per group of claims, plus the bounded re-asks—and the naming pass is a single point of dependence for the whole prompt. A group sees only its own claims, so anything that has to cross between groups must travel through the shared list of names rather than through another claim's wording. And every claim that degrades to per-claim verification is a claim whose plan was not used, which is why the accounting keeps those separate rather than folding them into the plan's measurements. Section 7 reports what this construction buys: higher recall than independent per-claim verification, at a slightly higher call count.

5.3 Validating a plan before it runs

Because the plan is code, it can be checked mechanically before any tool is invoked. Each line is parsed; a line that does not parse, or that is not of the form `check(cN, ...)`, is rejected rather than guessed at. Only calls from the vocabulary in table 5 are permitted, and each is checked for arity and argument shape—a count must wrap a `count` term, a spatial relation must name two objects and a relation from the fixed vocabulary, a directed motion relation must carry a target. Logical negation is accepted only directly around the three assertion-style operations, because a negated measurement would require uncertainty handling the downstream rules do not provide.

Failures are handled at two granularities, and neither is silent. If an entire line is unusable, the claim falls back to direct verification by the general verifier, and that fallback is *charged*—counted in the plan's cost like any other call, so that fallback never looks free. If a single term is unusable—an invented relation, a directed motion without a target—only that term degrades to a generalist judgment; the substitution is recorded and explicitly counted as *not* a measurement, so that a plan cannot inflate its measurement rate by mislabeling salvaged terms. Every plan carries this accounting with it: how many calls, which claims fell back, which terms were salvaged.

```
check(c0, judge("a dolphin leaps out of the water"))
check(c1, before(window("the dolphin leaps"), window("the dolphin dives back under")))
check(c2, judge("two salmon emerge at the same moment") and eq(count("salmon"), 2))
check(c3, judge("the salmon are behind the fishing boat") and rel("salmon||fishing boat", "behind"))
```

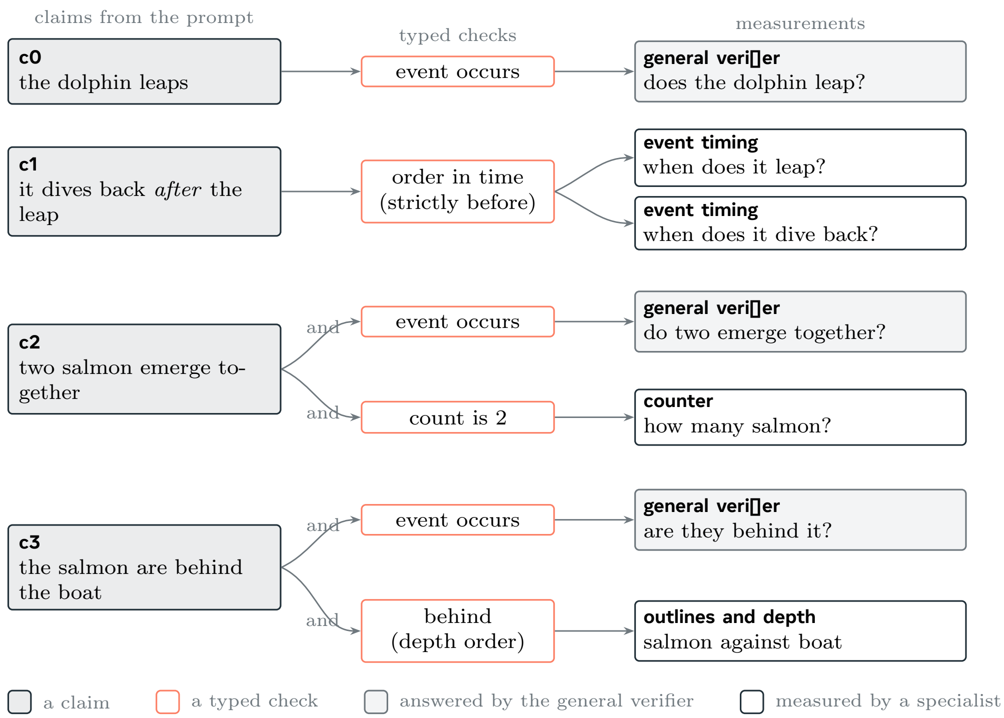


Figure 12 One prompt's plan as a graph. Each claim the prompt makes becomes one or more typed checks, and each check names the measurements that would settle it. A claim can carry more than one check—a numeral adds a count beside the occurrence judgement, a spatial preposition adds a depth-ordered relation—and a single check can consume more than one measurement, as an ordering does. Occurrence and existence go to the general verifier reading the frames; quantities, timings, outlines and depths come from specialists. Illustrative: the plan is written in the report's measurement vocabulary, and the claims are of the kinds the planner emits.

5.4 Several checks per claim, and negation

A claim rarely reduces to a single test (figure 12). Because an expression may join several calls, each conjunct becomes its own sub-claim with its own measurement, and the claim's outcome is composed from theirs by a fixed rule: **if any part is contradicted the claim is contradicted; if every part is supported the claim is supported; otherwise the claim is left unresolved**. Negation is applied at this composition step rather than pushed into the measurement layer, so the measurement rules never have to reason about negated queries.

Composition is what lets one sentence be checked in more than one way (figure 13). A claim that a ball triggers a cascade of a thousand mousetraps yields both a judgment that the cascade occurs *and* a count check against the number. Under the composition rule, a measured count below the stated number is by itself sufficient to contradict the claim even when the occurrence part is supported—a claim carrying only an occurrence check cannot express that.

5.5 How the video's timeline wires the plan together

The plan is not a list of independent questions answered in parallel and then tallied. What makes it a graph is that measurements are wired to one another through the clip's own timeline: one measurement decides whether a second runs, where in the clip it runs, and what its answer is permitted to mean. Figure 14 follows a single claim about order through that wiring on a real clip.

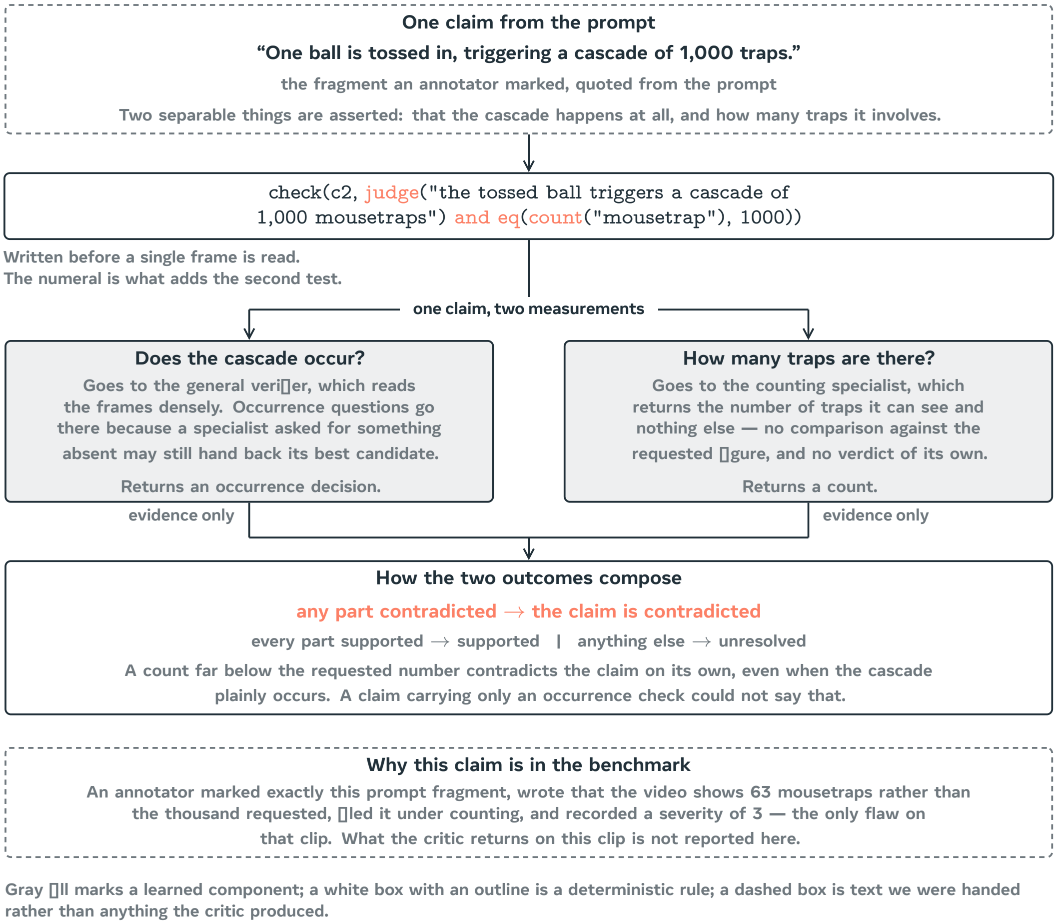


Figure 13 One claim, two measurements. A numeral in the claim adds a counting check beside the occurrence judgment, and the composition rule lets the count alone contradict the claim even when the cascade plainly happens. The prompt and the annotator's record shown at the foot are real; what the critic returns on this clip is not reported here.

One measurement gates another. A claim that one event precedes another is not settled by asking two timing questions and comparing the answers. The occurrence of each event is settled first, by the verifier that reads the frames densely, and only if both are confirmed are their measured intervals allowed to speak. If either event is judged not to occur, the ordered claim is contradicted outright and the timings are never consulted; if either cannot be placed in time, the claim is left unresolved. This ordering of operations is what keeps a returned interval from becoming backdoor evidence that an event happened—the failure mode of section 4.4, where a timing model asked about an absent event returns a confident interval anyway. Timings compose an order; they never establish existence.

One measurement scopes another. Several measurements are far too expensive to run over every frame—a depth comparison between two objects costs tens of seconds per frame, so a single spatial question over a long clip could cost an hour. Rather than sample frames, which is how a brief event gets missed, the plan asks a cheap question first: the whole clip goes to the verifier as frames labelled with their timestamps, and it is asked only *when* the surface or the pair of objects is on screen. That answer is not a verdict and is never treated as one—if it says the objects never appear together, that claim is ignored and the whole clip is measured. What it produces is an interval, which is padded, floored to a minimum length, and turned into an explicit list of frame times handed to the expensive measurement, which then reads *every* frame in that list. The saving is only the frames outside the interval, and a dozen enumerated conditions—an error, an unparsable answer, an interval covering most of the clip, a claim about absence, an unreadable frame clock—fall back to reading the whole clip. Every result records which of the two happened, so a scoped measurement is never mistaken for a whole-clip one.

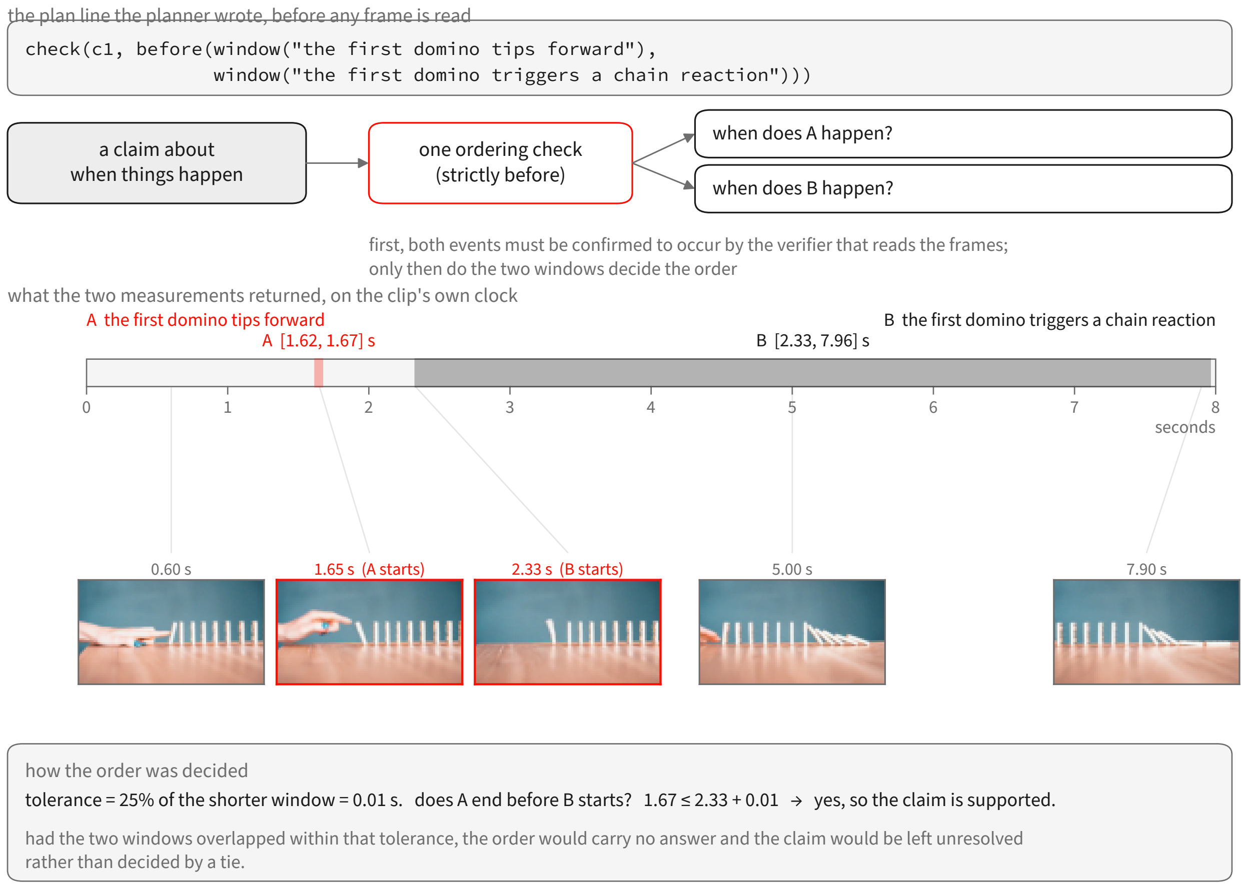


Figure 14 One claim about order, followed through the plan on a real clip. The planner writes the line before any frame is read. The ordering check requests two event timings, but they are consulted only after the verifier that reads the frames confirms both events occur. The two intervals returned are shown on the clip's own clock, with frames decoded at the instants they name: the first domino tips at 1.62–1.67 s, the chain reaction runs from 2.33 s to the end. The order is then decided by arithmetic with a tolerance of a quarter of the shorter interval. Every number and every frame here is from the recorded run.

The composition is arithmetic, and it can decline. Once two intervals exist, the order is decided by a rule with a tolerance rather than a threshold on raw numbers: the slack is a quarter of the shorter of the two intervals, so "before" requires a gap the clip can actually resolve. Two intervals that begin at the same instant, or that each span the whole clip, carry no order, and the claim is left unresolved rather than decided by whichever number happens to be smaller. The figure shows the arithmetic as the compiler applied it.

5.6 Routing by what the claim says

Which measurement a claim receives is decided by its content rather than by a blanket rule. A numeral introduces a count check alongside the occurrence judgment; spatial prepositions introduce a spatial relation; directed or shaped motion introduces a trajectory check; ordering words introduce a temporal relation; a described sound introduces the audio operation. Anything else—and anything the planner is unsure about—is sent to the general verifier. That fallback guarantees the claim remains executable; it does not guarantee a correct verdict. Relations outside the fixed vocabularies are never invented; they fall back to a general judgment.

Routing interacts with a property of the specialists that we treat as load-bearing: **a specialist's output is not by itself evidence that the queried thing exists**. Some specialist interfaces return their best candidate even when the query has no referent in the clip—a returned time window or bounding box can therefore reflect the query rather than the content. Existence and occurrence decisions are consequently assigned to the general verifier, which sees the frames densely, while the specialists supply the quantities, windows, masks, depths, and tracks it cannot produce. This assignment avoids treating a forced specialist output as proof of occurrence; it does not make absence judgments infallible. Composite claims combine the two: an ordering claim is first confirmed as two occurrences by the verifier, and only then ordered using the specialist's windows.

5.7 Executing equivalent measurements once

Sharing happens at two levels, and we keep them distinct. Within a plan, repeated references may share one internal description, so a prompt mentioning the same object in five claims does not describe it five times. At execution, two measurements collapse into a single tool call only when the complete measurement request is identical—the operation, the referent, the exact query, the output kind, and the time range—so a whole-clip measurement and a sub-interval measurement are never merged. This is exact-request reuse, equivalent to common-subexpression elimination in the database sense [41, 42]; no sharing across merely similar queries is performed. Whether this yields a net saving in practice is a question for the evaluation, not a claim made here. Temporal relations remain directional and tolerance-scaled, following a subset of Allen's interval algebra [1].

5.8 What the planner does not change

The planner is deliberately the only part that changes: a deterministic translator translates its lines into exactly the plan representation the compiler already consumes, so the compiler, the sharing rules, the measurement contracts, the execution layer, and the division between measuring and judging are untouched. This separation is intended to keep the planner replaceable and to prevent a change of planning style from silently altering how evidence is interpreted.

5.9 How often a plan asks for a measurement

Before considering verdicts, one property of a plan can be read off the plan itself: how often it asks for a measurement rather than a general judgment. This was checked with real model calls at the planning stage, without video or tools. A claim is *measurement-carrying* when its validated plan contains at least one non-generalist operation that survives validation without being salvaged or falling back; a claim combining a general judgment with a count counts once, and salvaged terms never count.

On 233 claims drawn from 22 human-annotated clips, every claim produced a valid, type-correct, compilable plan—no fallbacks, no schema errors, no compile errors—at one to two model calls per clip. Measurement-carrying claims were 27/233 (11.6%), against 2.3% for the earlier tabular format on the same claims: a roughly fivefold relative increase that nonetheless leaves a large majority of these claims resting on a general judgment, because these prompts are dominated by occurrence-style assertions. On two separate diagnostic sets chosen to be measurement-dense, the rate was 19/30 (63.3%) on claims drawn from counting, spatial, and ordering flaws, and 10/10 on synthetic counting prompts. Those diagnostic sets probe the routing rules; they are not an estimate of coverage or verdict accuracy on the evaluation benchmark.

Density of measurement is a property of the plan, not of the verdict it produces; its effect on what the critic catches is measured end-to-end in section 7.

Three gaps in the vocabulary are worth naming. The spatial vocabulary has no "on top of" relation, so such claims degrade to a general judgment; and negated measurements are unsupported by design. Out-of-vocabulary calls appear occasionally; the validator converts them into charged fallbacks rather than accepting them silently.

Where specialists run. Two configurations expose the same specialists by different routes. In the *routed* configuration each claim is dispatched to its dedicated specialist (table 4), which runs as an executable stage. In the *plan-based* configuration the compiler's typed checks call those specialists through a tool server, and a check reaches one only when the plan names a measurement; occurrence-style claims route to the dense-frame generalist by design, which is why measurements account for a minority of executed checks (section 7). Global plan selection among alternative tool recipes, and a value-of-information loop that spawns follow-up measurements, are not implemented.

6 The Intended Re[]nement Loop

The three components are meant to close a loop: the benchmark trains and evaluates the critic; the critic scores generated clips; and the critic's verdict becomes a reward that fine-tunes or guides the generator toward physical faithfulness. **This loop is future work.** The generator and the critic run on disjoint data—the generator on simulation-driven clips, the critic on the human-annotated benchmark—and no reward-driven generation run exists. Two prerequisites remain: a critic whose reward is trustworthy on the frozen human labels (section 7), and a decision about where the reward enters generation—preference optimization, guidance, or data selection.

7 Evaluation Protocol and Measurement

This section states how the critic is measured and reports what those measurements returned. A dash marks a measurement that has not been run.

Target and ground truth. The evaluation core of section 3, scored under the conventions of section 3.1. Human labels are the only ground truth.

Matching is outside the critic. To decide whether a critic output corresponds to a human-flagged flaw, a separate protocol uses two judge models (a GPT-family and a Gemini-family model, majority over runs). These judges perform *flaw matching only*; they are not components of the critic, do not produce its verdict, and are not treated as ground truth. Any "majority-of-three" in this project refers to such a model-vote protocol, never to human annotators.

Dense-frame requirement. The critic must observe the video densely. The evaluated components extract frames densely under a logged sampling policy, and any measurement obtained from a sparsely sampled input is excluded. A re-run counts only if it uses the dense path on the unchanged frozen target with BF16 tools at pinned revisions.

The comparison. The critic is compared against per-claim verification: the same claims, verified one at a time by the general verifier, with the question phrased as the critic phrases it, since the phrasing alone changes how readily a verifier answers. Because a detector call, a tracker call and a full vision-language judgement do not cost the same, cost is reported as model and tool calls per clip. Uncertainty is estimated by resampling whole clips, because flaws within a clip are not independent.

7.1 Results

Table 6 reports the paired comparison completed to date: the plan-based critic against per-claim verification, on the 149 clips of the evaluation core both systems have scored. Both receive the same claim texts and the same matching protocol, and the per-claim baseline is wording-controlled—asked its questions in the form the plan-based critic uses, because the phrasing alone changes how readily a verifier answers. Scoring follows section 3.1.

The plan-based critic catches **18 more flaws**, 6.3 points of recall, for about one extra call per clip. Resampling whole clips with a fixed seed puts that difference at 95% confidence in $[+8, +28]$, so it is a real difference on this set rather than sampling noise.

Two qualifications bound what the number means. It is a trade of accuracy against cost, not a dominance result: the plan-based path spends about 7% more calls per clip, and no run holds the two systems to an equal call budget. And this set is the set the system was developed against, so the figure describes fit to it rather than generalization.

System	Flaws caught of 286	Before collapsing duplicates, of 304	Allegations that landed	Calls per clip
Plan-based critic	**197 (68.9%)**	197 (64.8%)	40.1%	13.0
Per-claim verification, one claim at a time	179 (62.6%)	179 (58.9%)	40.2%	12.1

Table 6 Paired results on the 149 clips of the evaluation core that both systems completed; the one missing clip is an infrastructure loss. Those clips carry 304 of the core's 306 flaw entries, which collapse to 286 distinct spans (section 3.1); the second column repeats the same allegations against the uncollapsed flaw list. "Allegations that landed" is the share of each system's accusations that the matching protocol linked to a human flaw; it is reported because recall alone rewards over-flagging. Calls count model and tool invocations and exclude the matching protocol, which sits outside the critic.

Where the misses come from. Of the 89 flaws the plan-based critic missed, two were never decomposed—no claim in the plan covered them. Eight were flagged but not linked to the human flaw by the matching protocol. The remaining seventy-nine were checked and cleared: a claim covered the flaw and the critic judged the video fine, and seventy-six of those were decided by a yes-or-no judgement with no measurement attached rather than by a specialist returning a wrong number.

The bottleneck is therefore measurement coverage rather than decomposition or specialist accuracy. Of 1,917 executed checks, 1,414 ask whether an event occurs and 233 whether something exists, against 22 spatial relations, 19 trajectories, 18 orderings, 18 text reads and 16 counts: 14.1% are measurements. The claims this data produces rarely ask the plan to measure anything, which makes claim typing, not tool accuracy, the next lever.

Cost. Per-clip wall time over the 149 clips has a median of 102 s and a mean of 373 s; the ninetieth percentile is 1,087 s and the maximum 6,119 s. The mean sits far above the median because a minority of clips invoke dense per-frame specialists, and those clips set the wall-clock rather than the typical one.

What this does not cover. The paired set is 149 of the 150 benchmark clips. The cost-matched comparison in both directions and per-clip GPU-second accounting remain unrun, as does every flaw-level external comparison in table 8; the rating-level comparison of section 7.2 is separate and has been run.

7.2 Agreement with human ratings, against a purpose-trained judge

Flaw recall is not commensurate with what most published evaluators emit, which is a clip-level scalar rather than an allegation that can be matched to a localized human flaw. There is, however, common ground: both kinds of system can be asked to rate a clip on the scale a human used. VideoScore2 [20] is the natural opponent there—a judge trained for exactly this task, with a published evaluation script.

The comparison follows that script rather than a protocol of our own. On a sample of VideoPhy-2 [6] test clips, each judge emits an integer 1–5 per dimension; agreement with the human rating is scored as exact accuracy, accuracy within one point, and the linear and rank correlations (PLCC and SRCC), with unparsed predictions excluded pairwise. The sample is 200 clips drawn uniformly with a fixed seed, and the sample, the score definition and the metrics were registered before any result existed; nothing in the scoring function is fitted to these clips. Of the 200, 55 carry the benchmark's *hard* flag.

Table 7 reports both judges, and the two families of metric disagree: neither judge leads on both. VideoScore2 is better calibrated to the human's absolute point on the scale, winning exact agreement (33.0% against 31.0%) and agreement within one point (83.0% against 76.0%). The critic correlates better with the human ordering. Bootstrapping the paired per-clip difference over the same 200 clips, the rank-correlation advantage is $+0.138$ with a 95% interval of $[+0.005, +0.277]$, which excludes zero only narrowly; the linear-correlation advantage is $+0.110$ with an interval of $[-0.018, +0.242]$, which does *not* exclude zero. On the full sample, then, the ordering advantage is real but slight, and the linear one is not established.

	All 200 clips				Hard 55	
Judge	**exact**	± 1	**PLCC**	**SRCC**	**PLCC**	**SRCC**
VideoScore2 [20]	**33.0%**	**83.0%**	0.353	0.327	0.239	0.151
This work	31.0%	76.0%	**0.464**	**0.465**	**0.480**	**0.460**

Table 7 Prompt match: agreement with the human semantic-adherence rating on 200 pre-registered VideoPhy-2 test clips, scored with VideoScore2's own evaluation script. VideoPhy-2 also carries a physical-commonsense rating; the critic emits no score on that scale, so that dimension is outside this comparison.

Evaluator	**Output it produces**	**Caught**	**Recall**	**Calls**
This work, plan-based	localized allegation	197	68.9%	13.0
Per-claim verification	localized allegation	179	62.6%	12.1
Question-decomposition scoring [9, 22]	per-question answers	—	—	—
Alignment scoring [21, 31]	clip-level scalar	—	—	—
Learned quality scoring [19, 20]	clip-level scalar	—	—	—
Modular video question answering [10, 37]	per-question answers	—	—	—

Table 8 The intended external comparison, stated before it is run. The first two rows are the measured systems of table 6, on its 149 clips and 286 flaws. Dashes are unrun. A clip-level scalar has no allegation to match against a localized human flaw, so scoring it requires a stated rule for turning a score into a decision; that rule will be fixed in advance and reported with the result.

The hard slice is where the two separate. On those 55 clips the critic's advantage is $+0.242$ linear ($[+0.018, +0.511]$) and $+0.309$ rank ($[+0.050, +0.583]$), both excluding zero, as VideoScore2's rank correlation falls to 0.151 while the critic's holds at 0.460. That is the pattern one would expect if evidence assembled per claim degrades more gracefully on difficult clips than a single learned scalar, though 55 clips is a small sample and the intervals are correspondingly wide.

Why only one of the two scales is reported. VideoPhy-2 rates each clip on a second scale, physical commonsense, and the registered plan mapped the critic's output into that column as well. That analysis is not reported as a result, because it does not measure what the column is for. The critic's score is the fraction of a clip's claims that its checks contradicted, converted to a 1–5 number; rating how physically plausible a video looks is a different output from deciding whether a stated claim is contradicted, and the critic emits no score on the former scale. Comparing the one against the other measures the mismatch, not the capability. Producing such a rating would need a head that consumes the kinematic and trajectory checks directly, which does not exist here. This analysis therefore provides no evidence about physical-commonsense rating, and none about physics-flaw detection either; the flaw-level evidence for that is table 6, on human-localized flaws.

Two further limits. This is one fixed sample of 200 clips, so it bounds agreement on that sample rather than across the benchmark. And the two judges are not cost-matched: the critic runs a decomposition and a set of tool calls per clip, VideoScore2 a single forward pass, so nothing here speaks to efficiency.

7.3 The intended flaw-level comparison

The rating comparison above does not settle the flaw-level question, because a clip-level scalar cannot be matched to a localized human flaw. Table 8 fixes that comparison, so that the protocol is stated before the numbers exist. Each will be scored on the same frozen set, against the same human labels, with the same matching protocol, and each will be given the same claim decomposition where it accepts one.

The set was inspected and used during optimization, the wording-controlled per-claim baseline was introduced after treatment outputs had been observed, and no retroactive held-out split exists within these 150 clips; a claim of generalization therefore requires evaluation on a new frozen set.

8 Related Work

8.1 Physics and structure in video generation

Physical-commonsense generation and its evaluation are studied by VideoPhy [5, 6], PhyGenBench [36], and Physics-IQ [38]; V-JEPA-style work argues physics is best judged in a learned state space [14]. Our generator does not learn physics: it *imports* it from a simulator and asks the backbone to render it, via structure conditioning. Structure- and motion-conditioned synthesis has a strong lineage—ControlNet for images [57], motion control for video [49], the Wan family and its VACE control interface [25, 48], and, as guided editing rather than control-branch conditioning, SDEdit [35]. DiffPhy conditions a diffusion backbone on a physics chain-of-thought [56]; we instead condition on a rendered depth field from a simulation.

8.2 Evaluating text-to-visual faithfulness

Decomposing a prompt into checkable pieces is established: TIFA [22] and the Davidsonian Scene Graph [9] score generated questions/propositions with a vision-language model, VQAScore uses an image-to-text model [31], and GenAI-Bench, VBench, T2V-CompBench, FETV, and EvalCrafter benchmark compositional alignment [23, 28, 32, 33, 45]; embedding and preference scorers predict alignment or human preference [21, 27, 53], and learned quality models predict human judgment directly [19, 20]. We keep such a holistic learned scorer out of the verdict path. Relative to these, our critic routes each claim to a dedicated open perception specialist under an evidence-only contract, validates a typed plan before execution, distinguishes event occurrences, reuses perception only under an exact contract, and preserves claim-level provenance with an explicit **unknown** outcome. We claim novelty for the integration, not the parts.

8.3 Visual programs, tool-use, and modular video QA

Composing perception modules for reasoning includes Neural Module Networks [4], VisProg [17], ViperGPT [46], CodeVQA [44], and neural-symbolic VQA [24, 55]; LLM tool-controllers and plan-execute graphs include HuggingGPT, TaskMatrix.AI, ControlLLM, LLMCompiler, and ReWOO [26, 30, 34, 43, 52], with search-based variants [7, 54, 60, 61]; modular video QA includes ProViQ, MoReVQA, the VideoAgent systems, and VideoTree [10, 12, 37, 50, 51]. Our execution layer is narrower and verification-specific: a *typed* plan, type-checked before tools run, whose evidence obligations carry three-valued semantics; it does not search over alternative plans. Reuse of equivalent measurements is common-subexpression elimination in the multi-query-optimization sense [15, 16, 41, 42], and the specialists themselves draw on open detection, counting, tracking, grounding, and audio models [2, 3, 8, 11, 13, 18, 29, 40, 58, 59].

8.4 Flaw and preference datasets

Prior video benchmarks largely provide clip-level physical-commonsense or preference labels [5, 6, 36]. Our benchmark instead provides *localized* human flaw annotations—severity, the violated prompt fragment, a rationale, and, where present, a time span and a box track—so that a detector can be scored not only on whether a clip is flawed but on whether it found the specific, spatially and temporally located error a person marked.

9 Limitations

Generation. The control signal comes from a simulator, so scene diversity is bounded by what is simulated, and the mapping from simulation to prompt appearance is not itself verified for physical consistency.

Benchmark. The corpus is recall-only (no clean clips), so precision is not directly measurable; flaws are single-annotator with no inter-annotator agreement; the scope/category tags are machine-derived; and the source generator of the clips is unrecorded. Repeatedly designing against one frozen set risks overfitting to it even without training on it.

Critic. The spatial relation predicate and negative-event verification are limited; the typed-plan layer does not yet dispatch the full specialist registry; binding remains a central failure mode, since the compiler enforces symbolic identifier consistency but cannot guarantee that heterogeneous tools localized the same physical instance; and the reasoner-mediated sub-checks reduce but do not remove semantic opacity.

10 Conclusion

Physical faithfulness in generated video requires steering, measurement, and ground truth together. This report contributes one piece for each: a generator that imports geometry and motion from a simulation and renders them through a video-diffusion control branch; a human-annotated benchmark that localizes real flaws in space, time, and prompt reference; and an auditable multi-tool critic that decomposes a prompt, routes each claim to an open specialist returning evidence only, and combines the evidence deterministically into a three-valued verdict. Against human labels, the compiled-plan critic recovers more localized flaws than verifying each claim on its own, at a slightly higher call count; an equal-cost comparison and a held-out set remain to be run. Closing the loop that lets the critic refine the generator is the next milestone.

A Appendix

To be expanded: the full generation configuration and control-render details; the benchmark schema and per-stratum statistics; the critic's plan schema, predicate type rules, and temporal-relation truth tables; the specialist license and revision manifest; and the pre-registered evaluation manifest.

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